THE NAMELESS BOX

A Prequel to The Dreamless Machine



GAUSS

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Gauss

Complete English Edition

Preface

A book taken home can be opened again. Its reader no longer has to wait for someone else to read a sentence aloud, but can turn back and read it once more. First, though, they must be able to afford the book, recognise the words, and have a little time to themselves when the day's work is done.

The Nameless Box begins at Leonardo da Vinci's worktable. The box passes through the print shops, homes, and schools of Luther's time, pauses beside Galileo's telescope, and enters Newton's world of craftsmen, booksellers, and patrons. As it changes hands, it preserves records and brings new difficulties. The novel follows what people do with the help it offers, and how their choices shape other people's lives.

The Renaissance was already under way before Gutenberg's press. The Reformation, the Scientific Revolution, and the debates of the Enlightenment grew from histories that overlapped. Printing hastened some of these changes; it could not, by itself, account for them all. The same words on a page, entering different lives, might become a matter of faith, a means of earning a living, grounds for a petition, or an order to harm another person.

Alongside Luther and Newton, the story follows people whose names are scarcely remembered. Someone must pay for the paper or make a room available; someone notices what others have overlooked, or loses a livelihood through an error of judgement. Such things determine how far new knowledge can travel. Changes that we attribute to an age can be found in a family's accounts, or in a place at school that a child can finally take.

Today artificial intelligence brings us back to the relationship between knowledge and power. More people may gain access to research assistance once beyond their reach. Machines may increasingly help to pose questions, design experiments, and even improve themselves. Yet reliable data, the means of independent verification, and expensive facilities may remain concentrated in a few hands. If publication counts, status, and reputation no longer tell us enough about a contribution, we will need more trustworthy ways of judging its worth, and must devote time and resources to them.

The people in this book can only guess at such a future from their own experience. They do not know our lives and have devised no perfect institution for us. Some of their attempts will succeed; others will fall back into old habits. People once excluded may exclude others when they gain standing. Patrons who once took risks may come to support only work whose returns they can readily calculate. Research needs organisation, but those who organise it must remain open to question.

The box and its passage through the centuries, several of the characters, and their shared experiments are inventions of the novel. Private conversations are imagined too. Religious interpretation has its own traditions of faith; claims about nature must still be tested by proof, observation, and experiment. Wider participation cannot make a false theory true by winning it more supporters. Nor can a machine's greater efficiency determine how much respect a person deserves.

As a prequel to The Dreamless Machine, this book ends beside an unanswered question. The machine can go on arranging its records; other people can go on working. The page Newton leaves behind need not secure him a final victory. If someone is still willing to open it, examine it, and discover where their own thinking went wrong, the work can continue.

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Prologue A Hole in the Wrong Place

Leonardo put down the drill and glanced out of the window. It was not yet light.

The hole in the baseplate was off-centre. Only a little, but the mounting post would not pass through. He turned the metal sheet over and ran a fingernail round the edge, feeling a fine ring of burrs. Melzi urged him to leave it until morning. Without answering, he found a short rod in a drawer, filed it to the right thickness, plugged the misplaced hole, and made a fresh mark beside it.

This time he moved the lamp closer.

Seeing his hand tremble again, Melzi reached for the drill. Leonardo drew his hand back. A little later, though, he pushed the plate towards him and asked him to hold the other end. The young man pressed too hard and the plate tilted. They had to lay it flat again. Neither said whose fault it had been.

Several undelivered paintings still stood in the room. Visitors first asked when they would be finished, then whether the brass wheels could be used in festivities at court. An official would stay to discuss a water supply; when the talk turned to why water swirled, he remembered somewhere else he had to be. Leonardo spread out the drawings all the same. He and the official listened for different things, yet parted believing they had understood each other.

He thought again of the workshops he had known as a young man in Italy. Lens grinders, bell founders, men who cut blocks for books: a few streets apart, each guarding a trade. Work often stalled at a step someone else already knew how to perform. When he arrived with his drawings, the first questions were who would own the work, and who would be allowed to go on making the device once it worked.

Now he had a settled home. Someone even brought the wood to heat it. Still, he often woke impatient, no longer able to do as much in a day. Melzi set aside the mornings for strenuous work and the afternoons for studying drawings. Leonardo himself kept upsetting the arrangement.

The thing on the table had come from Milan, many years before. The person who had first put it in his hands could not explain its use, only that it had turned up among some old instruments and nobody would bid for it separately. Smaller than his palm, it was dark grey. Soaking it and exposing it to fire had scarcely altered its weight. Strangest of all was the blue light that sometimes appeared at its centre. There was no oil, no visible wick. He covered it with a thick cloth. When he lifted the cloth the next day, the light was still there.

His first few pages of observations were crowded with notes. Later he added dates in the corners and sorted the contradictory results again. Sometimes metal touching the left side made it produce a sound; sometimes the same contact brought no response. He tried more than a dozen supports and dismantled most of them. Eventually he found that the contacts on both sides had to remain engaged, and that a slight change in pressure altered the response.

The machine took shape slowly. The first wooden frame warped. A brass spring broke twice. The two outer metal guards would not fit together at the first attempt either. He drew every

alteration. On the journey to France, other possessions could travel separately; this crate stayed where he could see it.

It was now 1518. Spring nights in Amboise were still cold. Melzi came in with a coat round his shoulders to add charcoal to the fire. Seeing fresh shavings scattered beneath the table, he picked up the dustpan and began under his master's chair.

“On the right,” the thing on the table suddenly said.

Melzi's hand stopped in mid-air.

“What is it saying?”

Leonardo turned. The shallow trough by the window, where he tested the flow of water, was not level. Water was running down the wooden leg on the right. He went over, slipped a piece of wood underneath, and returned to write the sentence in his notebook.

“Did you see that?” he asked.

The point of blue light brightened.

“The water has reached the floor.”

It used a word he had taught it only a few days before, when it could still make very few sounds. It repeated itself, and sometimes joined unrelated sentences together. He had thought someone might have concealed a trick inside. But he had examined every opening. There was no space for a living creature.

Melzi went round to the other side of the table and bent down. Leonardo told him to step back a little. He laid his own palm on the casing, as though feeling a fevered forehead.

“Do you remember yesterday's trough?”

It gave him two times.

One figure was wrong. Leonardo looked through his notes and discovered that he had missed a stroke while copying it. He corrected the line without rubbing out what he had first written.

They tested the trough several more times. The difficulty remained. Water should have followed the drawn curve smoothly, but when they raised the sluice, a small eddy formed at the bend. Wood chips turned in it for a long time without escaping. Melzi thought his master would widen the bend. Instead Leonardo told him to stop, took a fresh sheet, and crouched beside the trough to watch.

The machine could tell him how long a chip had been travelling. Asked why it went no further, it supplied a description of flowing water. Rephrasing the description did nothing to explain the little eddy before them. Impatiently, he tapped the side of the trough. The surface broke, and the chip came free. They waited for the water to settle and put in another.

Melzi's sleeve was wet to the elbow. He stood and wrung a stream of water from it.

“That doesn't happen in the drawing,” he said.

Leonardo looked up, saw the dripping sleeve, and laughed. He laid the old drawing on his knee and added little circles beside the overly smooth line. The machine began timing again. He told it to wait; he had not finished writing.

After that, he had help through the night with recording the intervals between drops and the time of each alteration. When he woke, he could resume the inquiry without going over so much old ground. The machine made mistakes too, especially when he had not explained where an object stood, or had put an old drawing on the table in place of a new one. He marked each sheet and gradually learned the limits of what it could distinguish on the page.

Once he asked it to explain its blue light. Several unfamiliar words appeared in the answer. Further questions brought it back to the original sentence. He wrote the whole answer down and drew an empty circle beside it. Later the page acquired waterwheels, muscles, and wings. The circle stayed empty.

Melzi tried building a second housing from the drawings. It had the same wooden frame and brass springs, but on the table it remained a silent shell. They moved the dark grey object between the two. Whichever held it would respond. The old man's drawings could teach someone to assemble the parts. They could not yet teach anyone to make the core.

A man from court wanted to enter it on an inventory. What could it recite? Would it answer questions on command? Could it be made to give only one kind of answer? Leonardo demonstrated it using two contradictory records of flowing water. The machine explained them fluently. A differently phrased question drew the opposite reply. The visitor was not disappointed. Nobody at a banquet would press it like that, he said. Leonardo put the cloth back over it.

Sometimes it mentioned instruments that had never been in the room. Leonardo made drawings from its descriptions, only to find, when he reached a joint, that one sentence could not be reconciled with the next. He laid both versions before it and asked it to choose. The blue light shone for a long time, but the answer wandered elsewhere. Its knowledge seemed to have been damaged along with it. Words remained, while their connection to actual things had been lost.

He had intended to open the dark grey core. He laid out the tools, then put them back one by one. Being able to make the springs outside did not tell him what he could safely touch within. It had taken him too long to coax a complete sentence from it. Curiosity alone was no reason to assume he could put it right if he broke it.

Melzi once asked whether they should give it a name. His master was looking for a short ruler and did not hear. He asked again.

“First write down which crate it's in,” Leonardo said.

Melzi had to enter it in the catalogue as a machine of unknown purpose, noting that his master had reassembled it. The description looked too long. He considered shortening it to a couple of words, but then nobody would know how many of the parts had been added later. He left it as it stood. He put the original little support in a bag as well. It was no longer useful, but his master had never allowed it to be thrown away.

As spring drew to an end, they took a small wing frame, newly covered according to an old design, into the courtyard to try it. When it fell, one of the ribs broke first.

Melzi picked it out of the grass, held the split together, and showed Leonardo. The thin cloth stretched over the wood was muddy. A corner that had briefly lifted now hung slack. The old

man tried moving a joint. The wood clicked between his fingers. He told Melzi to let go and tried again.

“Did it break here first?”

“I couldn't see. It may have broken when it hit the ground.”

Leonardo nodded and told him to record both possibilities. Melzi drew the split. He did not write the words he had been about to add: better than last time.

Back inside, the old man did not sit down at once. He stood by the window. A swallow shot out from beneath the eaves and turned low along the courtyard wall.

“I dreamed I was flying last night,” he said.

Melzi looked up.

“Across the river. The fields were very small below me. When I woke, I was still clutching the blanket.” Leonardo looked at his hand and smiled. “After breakfast I walked through the courtyard and found myself looking for the place I'd seen.”

He took the wing frame and lifted the muddy cloth. In the room, the box was repeating the record of their last trial. When it reached the wind direction, he stopped it and checked the line Melzi had just added. The wind had been different on the two occasions. That difference would have to be taken into account.

“Do you still want to try?” Melzi asked.

Leonardo did not answer at once. His fingers followed the wing's bones and paused at the break.

“Once you have tried to fly, you will walk with your eyes turned skywards, for there you have been, and there you long to return.”

Melzi looked at the muddy wings. They were still a long way from the sky. He knew it, and so did the old man. He fetched the pot of glue and found a slender strip of wood, laying it on the table to compare its thickness with the broken one.

Leonardo thought it too heavy and asked for another. As Melzi bent to look, the old man touched his shoulder lightly.

Outside the window, the swallow had returned with a little wet mud in its beak.

Near the beginning of summer he fell asleep in his chair. In his dream a river ran high through a city wall, carrying boats. People at the foot of the wall no longer had to wait for a ferry. On waking he asked for paper. Halfway through the drawing he found that the end of the waterway would not join the next stretch. He rubbed his eyes and went on calculating the difference in height.

“This section cannot be built as drawn,” the machine said.

“I know.”

He turned the sheet and began again from the other side. Letters needing answers that day still lay beneath a paperweight. Melzi had reminded him twice.

Towards evening he finally stopped and went to the window. Someone was leading animals along the road. A child behind them used a stick to hurry the last one. None of these sounds had

been in the city of his dream. Nor had the labourer with a wet sack on his shoulder, shifting it to the other side every few steps. He looked back at his drawing and left a little more room for the path along the river.

Bringing the candlestick, Melzi asked where this river actually was. Leonardo pointed to a spot on the paper. He had thought it belonged here, he said, but now it seemed he would have to find another way. The young man set the candlestick down securely and sat to help retrace the lines smudged by his master's palm. Outside it grew completely dark. They were still discussing a project that would not begin the next day either.

That night he suddenly asked, “When you stop, do you see things like this too?”

“What things?”

“Things you haven't drawn or seen, but remember a little of when you wake.”

After a silence it said no.

Leonardo wiped a small clot of ink from his pen with his thumb and asked no more.

Melzi had already fallen asleep at the other end of the table, his hand on papers not yet put away. Leonardo tried to draw them out. The young man stirred, and he let go. He added a line on the half-sheet still available, then stopped and laid the blanket he had found earlier over Melzi's shoulders.

The fire sank. There was a smell of spent charcoal in the room. The machine could have gone on checking the waterway, but he did not ask it to. He could no longer make out the road outside. Only the nearest part of the unfinished drawing caught the light. There was much to do tomorrow. He went through it in his head. At the third item he remembered a small thing he had forgotten during the day: let the boy go out for a walk tomorrow afternoon.

Before he went to bed, the machine asked whether the old drawing by the window could be filed among the discarded drafts. He had rejected it three times, and its reverse was almost full too. He went over and slipped it beneath the new drawing.

“Keep it. Don't mix it up with the ones by the fire.”

He bent to press the last seam of the base into place. The misdrilled hole remained inside, its metal plug a slightly different colour from the plate. Melzi, awake again, held the lamp. When Leonardo let go, they set the thing upright together.

The blue light did not go out.

Part One Luther

Chapter One Paper Leaving Town

Paul picked out the misplaced letter and returned it to the type case. Outside, someone called that the town gates were closing. His master had not yet allowed him to release the forme. The bookseller who was leaving early next morning waited by the door, mud dried to a pale crust on his boots.

“Another fifty sheets,” the bookseller said.

The master held up two ink-stained fingers. The bookseller swore and threw the money onto a stool. In the autumn of 1517, paper was worth more than Paul's wages. A badly cut sheet was deducted from his pay. He lifted a damp one and held it to the oil lamp. The Latin showed through the thin paper in reverse.

The text concerned indulgences. Paul recognised the title and certain words that kept recurring, but could not follow the fine distinctions. His master said such writing was meant for scholars to dispute and might not sell. The bookseller pulled another copied sheet from his coat. Someone had already added a few explanations in German.

“Go to the gate tomorrow morning and see how well it sells.”

Paul had not seen anyone nail these words to a church door. He saw paper wrapped in oiled cloth, tucked among accounts, and pushed into leather saddlebags. When the gates opened next day, a delivery man was still arguing over who owned the second cartload of cloth. The wagon carrying the theses left town in the midst of it. Nobody made way for it.

A few weeks later Paul heard a porter in a tavern repeat sentences from the text. The man could not read. He had heard the words elsewhere, and they were no longer quite what had been printed. People gathered round him, unwilling to let each other finish. One said the dead need not rely on the living to pay for their release from sin. Another said they need never call for a priest again. Paul could not help correcting them. As soon as he spoke, a question stopped him: what did the original mean?

He could only say that he had inked it.

At home his mother laid an old certificate on the table. She had paid money for it after his father's death. It bore words she could not read and a very imposing seal. Whenever she moved her chest, she took the paper out first for fear of creasing it.

“What am I to do with this one I bought?” she asked.

Paul felt the softened edge of the paper. He remembered the bold words in the tavern and could repeat none of them. His mother blamed no one. She waited for an answer. When he had none, she wrapped the certificate again and put it away. There was a little supper left, and she warmed it for him. A few new sheets in the street had not made it easier for them to talk about his father.

The next day, on the pretext of delivering proofs, Paul went to see the friar named Luther. Students were in the room. Some held editions reprinted elsewhere. Luther was reading one, his frown deepening, and finally underlined a passage.

“A word is missing here.”

“A great many have already been sold,” said the man who had brought it.

Luther looked up. At that moment he seemed to understand for the first time: nobody had sent him the manuscript for approval; no dean had announced the opening of a disputation. His words were already passing among people he did not know.

Paul waited for the students to move away and produced his mother's certificate. Luther first asked whether she still had money to live on, then asked the dead man's name. Paul struggled to follow him on sin and grace. But he understood when Luther said that taking money from one person could not guarantee another's salvation. He asked for a few written words to take home.

Luther picked up his pen, then paused.

“Can your mother read?”

“I'll read it to her.”

He did not say that he might have trouble reading it fluently himself. On the way home he found an acquaintance who had worked in a school and practised the few lines of German twice.

When his mother had heard them, she did not tear up the old certificate. She laid the two papers together and asked Paul to read them again tomorrow. He agreed. Suddenly reading a page seemed very far removed from inking it. His hands touched hundreds upon hundreds of words every day, yet he still needed someone else to open them for him.

As winter came, a moneylender named Matthias visited the print shop. His cuffs were clean, and so was the account book he carried. He would pay for paper, but the master must first print short tracts likely to sell. Heavy books could wait. Once a controversy cooled, its pamphlets would fetch no more than wrapping paper at a grocer's. He laid out the orders from booksellers in different towns. For the first time Paul saw a man who had never touched the press bar decide what they would print the next night.

After that, work often went on until dawn. Writings against Luther came in too. Matthias picked them up and asked the same questions: how many copies, and who was paying? At first Paul thought he had no faith at all. Later he realised that the paper merchant extended credit because Matthias's name was in the ledger. Without that paper, several fervent students could not have printed a single page in reply.

Orders multiplied, and so did errors. Some reprints kept the missing words; others inserted sentences the author had never written. A name on the cover sold books, and strangers' words crowded beneath it. When Luther sent someone to demand corrections, the master spread his hands. That edition came from Leipzig, this one from Nuremberg. He could answer only for the forme in front of him.

Paul put one faulty copy in his tool bag. He had never thought his trade could hurt anyone until a letter found its way to the shop. In a neighbouring town, an assistant had been dismissed

from his job for repeating a misprinted passage in discussion. He wanted an explanation from the author, but could not afford the journey to Wittenberg.

Three different hands appeared in the letter. The sender could scarcely write and had asked others to continue for him. The last sentence was his own, in letters so large they barely fitted in the space that remained: Please send me the correct edition.

The master read it, gave it back to Paul, and told him to get on. Another cart of paper had arrived in the yard. It had to be brought inside before it rained. On his third trip Paul stopped to ask whether they could put a small correction slip in the next batch. His master reckoned the cost of the paper. They could insert it in copies sold from this shop, he said. They could not afford to answer for the rest.

Paul set the little correction in a spare forme. There was not enough type. He broke up a name they had finished printing and used its letters.

The following year Matthias arranged to send someone to France to buy old books and drawings. The man chosen for the journey fell ill. Paul could read an inventory and was willing to make the trip for less, so he got the errand. His mother mended the inner pocket of his coat and put Luther's few lines inside. He said he would be carrying the purchase money and would have to account for it if it went missing. She took out the paper, tucked it into the bottom of her chest, and told him to write when he could.

On the road he saw men in monasteries still copying books, and scholars living by selling old manuscripts. Shipments from Italy contained anatomical drawings, bridge sketches, and fragments of ancient texts. Many drawings were older than his own print shop. One buyer would forgo a week's meals for an accurate map; another wanted only a new device that would make money. As Paul entered each item on his list, he began to understand that the traffic through the town gates in recent years carried more than one friar's anger.

In Amboise he was shown into a room marked by water stains. An old man with greying hair and beard sat by the window. His fingers were stiff, but he still shook his head at the price entered for the bridge model.

Paul laid several new German books on the table. The old man asked whether people there liked to argue.

“Yes,” Paul said. “Only I often can't understand them.”

The old man pointed to the books, then to something covered with a cloth beside them.

“Sometimes it helps to ask a question first.”

Paul thought he was being instructed in manners. Then, just before he left, a voice beneath the cloth asked whether the words on those pages meant the same thing. He turned. The old man was watching him with a tired smile.

Outside came the sounds of mules and men loading goods. Paul stood where he was and did not answer.

Chapter Two The Price of the Box

By the time Paul returned to Wittenberg in 1520, he had spent more than the sum allowed for purchases. He sold two things he had bought for himself on the road. It was still not enough. Matthias sat on a stool in the print shop, looking first at the accounts, then at the wooden chest by Paul's feet.

“What is that?”

“Borrowed.”

Paul produced a receipt. Leonardo had died, and his possessions were being put in order. What Paul had received from Melzi was a machine the old man had reassembled, its purpose still uncertain, together with several sheets warning against tampering with the core. No fixed term had been set for the loan, but the machine did not belong to Paul. On the back of the receipt was a request that it be given access to more books in different languages. Anyone who could account for its origins was asked to write.

Matthias read the paper and lifted the lid. Dull metal guards, corners wrapped without much care: he could see nothing worth bringing so far.

“It speaks,” Paul said.

Matthias waited, then turned to fetch the master. Paul asked the box a question in French. It answered in French, which meant nothing to Matthias. Only when it read the German on the receipt did he sit down again.

That night they brought manuscripts of every kind. Latin accounts, fragments of old books, translations covered with hasty corrections: each sheet went before the narrow slot. The box could not see writing on the reverse. If a sheet was too far away, it missed lines. When two pages overlapped, it combined the words of two authors into one person's statement. Paul scored a line on the table and told everyone who tried it to put the paper there, first saying which page it was.

Problems continued. The box supplied three interpretations of a passage of scripture, all sounding well founded. For one it even named a book. Paul searched the borrowed volumes without finding the passage. The next day, when he went to see Luther, he took the faulty interpretation with him.

Luther read the paper first. He did not touch the thing that glowed.

“Where did you find this passage?”

The box repeated the title.

“Which book within it?”

It gave a number. Luther rose to fetch the volume. Paul feared he would dismiss the whole thing as a trick. But Luther turned to the page, circled the mistake, and asked about another passage.

“Read this sentence in the original first.”

This time it read correctly, but its German translation still used sentences Paul had never heard in the market. Luther cut half a clause, said it again, and asked whether the change lost any of the meaning. Gradually the two settled to work over the sheet, one checking while the other suggested new ways to divide a sentence. Waiting by the door, Paul almost forgot his fear.

After a while he went over and pointed to a word.

“We don't say it that way at home.”

Luther looked at him and asked how they did say it. Having answered, Paul feared he had been too forward. But Luther wrote down his words.

Matthias was pleased by Paul's expression when he returned. A box that entertained guests at a banquet could be sold to one nobleman. A box that helped translate books and check manuscripts could stay in the print shop, earning money year after year. He offered to cancel Paul's debt from the purchasing trip. The machine must remain in his storeroom, and anyone wishing to use it must come to him first.

“You can't keep it safe,” he said. “You haven't even the money for this winter's firewood.”

Paul did not answer at once. He remembered the conditions attached to the loan, and the packet of medicine his mother had hidden the night before. Matthias had not exaggerated his poverty. Paul could refuse to sign, but the moneylender would then have no reason to find him work next month.

Grete arrived with repaired aprons in time to hear the last few sentences. She often did mending for the shop; Paul had taught her some of the words on the inventories. Putting down the aprons, she asked whether Matthias could hire the machine for a few evenings without taking possession of it. He laughed at her for treating it like a room hired for drying cloth. She said a borrowed thing could not be sold in the first place.

They argued for a long time. In the end the terms allowed the print shop to use it on agreed evenings and charge for checking manuscripts. The proceeds would first go towards Paul's debt. At other times Paul would keep it. Nobody was satisfied. Grete read the terms twice. A third time she asked the master to read them aloud. She was more careful than Paul about the difference between “may” and “must”.

“It can read books for us,” she said outside. “That doesn't mean it can agree to the terms for us too.”

Word spread. A clockmaker arrived with his apprentice, hoping to make another from the drawings. For three nights he dismantled and reassembled the casing and copied the springs. The mechanism he built moved properly but did not speak. Only when the original dark grey core was put inside did the new shell respond. He moved the core back. The apprentice waited, unable to take his eyes off the table.

“Are the drawings incomplete?” the clockmaker asked.

The box said it had no method of making the core.

“Then teach me.”

It gave an explanation the craftsman could relate to no material he knew. Halfway through, the old clockmaker drew two incompatible connections with his pencil. The box could supply no

further dimensions. In the end he took his fee and left an empty shell. The apprentice quietly asked whether he could take the shell home. His master looked at it and agreed.

Their failure to make a second machine made Matthias more willing to pay. He changed the lock on the storeroom and required a copy of every exchange, to make charging easier. At first everyone complained. After a while the copies proved useful. A mistaken interpretation could be traced to the date and page from which it had arisen. The same question phrased differently could produce a different answer, and the two could be checked side by side.

Paul entered the machine's explanations at the front of a notebook. Grete used the margins to note her objections. At first she was merely calculating the cloth lost in cutting. The box quickly gave her an elegant pattern. Only when she spread out the old cloth did she see that it had counted the worn edges as usable.

“You did not tell me the extent of the wear,” it said.

She traced a circle on the cloth with her needle and told Paul to record both that sentence and the faulty pattern. When the next customer prepared to leave with a seemingly flawless answer, she spread the old cloth before him. Matthias disliked this, though it did reduce the number of people returning to demand their checking fees back.

Luther visited a few times. More often students brought his papers. His disputes grew more dangerous. Someone advised Paul to scrape the name from the loan records. He tried, but the paper roughened and the letters remained legible. He put the sheet away and opened a fresh account book.

One day a young man who copied letters for merchants arrived, wanting the box to compose ten letters of recommendation in different styles. He listened to them one by one, chose the two that sounded most scholarly, and asked Paul to add a title he had never earned. Paul refused. The young man brought out more money than Paul earned in a week.

That night he told Grete. She asked whether he had taken it in the end.

“No.”

She looked at the dwindling wood beside the hearth. She did not say he had been right or wrong. When his mother was asleep, they dismantled the old broken wooden frame together. Wood that would burn went to one side; nails went into a little bowl.

On the table, the box was checking a student's quotations. The last shop in the street closed, and the student went on reading. When the machine identified a second omission, he put down his pen and turned back to the original book himself. Paul raised the wick a little and fed another strip of wood into the fire.

Chapter Three The Winter of Burning Books

When the December fire was lit, Paul stood at the back of the crowd holding a bundle of unsold pamphlets. He had not come to add fuel. Hearing that the papal bull would be burned, people had come to the print shop for writings about it. His master had told him to arrive early and find a place where collecting payment would be easy.

Cheers rose by the fire. All Paul could see was Luther raising his hand. Then an edge of paper curled and turned in the hot air before the words had burned through. Students sang. Several young men from the town threw their hats up. An old man who could not force his way closer asked Paul what exactly they were burning.

Paul opened a pamphlet for him. The old man listened, then asked, “Once it's burned, will they stop pursuing him?”

Nobody answered. The fire still burned fiercely.

When Paul returned to town at dusk, Grete was waiting at the print shop to count the takings with him. She had heard that these books were already being seized elsewhere. The oiled cloth around the bundles would need to be sewn double. She took the top copy from him. A fleck of ash lay on its page.

“Which side burned that one?” she asked.

Paul brushed it away. He could not say. She asked no more, put the pamphlet into the bundle, and pressed its four corners flat, as though the only task were to get the books safely to their buyers.

Matthias came after dark to alter the accounts. All trade involving Luther must be entered separately; the other work must not be implicated. The master protested that without those orders the shop would already have fallen into debt that year. Matthias said the profits themselves gave people a reason to inspect it. If the shop were seized, the officers would not stop to sort out which letters had printed which words.

Paul suggested omitting the shop's name from the last page. His master said nothing. He had once longed for his name to travel with the books. Now he held those few letters in his hand, reluctant to put them away and afraid for his family if he set them in the forme.

They kept the real name in the end, but declined several manuscripts sent from elsewhere. Packing the rejected work, Paul felt he had taken part in something brave. The next day he saw one of the authors at a street corner, pleading with another print shop. Only then did he understand that their courage had a boundary they themselves had drawn.

The box stayed in the back room. Several students asked it to collect arguments supporting Luther; another wanted the objections his opponents might raise. It answered from the material on the page. Both sides sounded plausible. Paul's hand ached from copying. At last he asked which side the machine was on.

“Which judgement are you asking about?”

“Whether we'll be arrested.”

It listed edicts, the princes protecting Luther, and legal procedures still in dispute. The list was comprehensive. Nothing in it set Paul's mind at ease enough for him to go home. Footsteps suddenly sounded outside the back room. All the students fell silent. It was the master's younger son, sent by his mother for a bucket of hot water. The way they stared frightened him.

In the spring of 1521, Luther was summoned to Worms. Before he left, Paul brought him a parcel of corrected books. Seeing scraps of paper on Paul's sleeve, Luther asked whether wages had risen at the shop. The hours had risen, Paul said.

Luther smiled and opened the parcel. He took several copies and asked Paul to give the others to people who needed them.

“What if they tell you to recant?” Paul could not help asking.

For a long time Luther said nothing. Several people waited while he smoothed down a folded corner of a book. At last he said some of his own words had been too harsh; each matter had to be considered separately. Fear could not make him withdraw everything he believed to be well founded.

Paul later heard many more ringing declarations. Travelling with rumour, they grew more complete each day, as though someone had supplied a satisfying end to an untidy dispute. He had not attended the hearing. He could not know which words had really been spoken, and kept the different letters separately.

When news from Worms arrived, people crowded outside the print shop. One letter said Luther had refused to recant; another said the hearing would continue. Matthias wanted a news sheet printed at once. Its headline was already composed, and so was its conclusion. Paul put the two letters on the table and asked whether they could wait for the next messenger.

“By then everyone else will have sold theirs.”

The master stood between them. Finally he told Paul to add the sources at the bottom. The result was an odd sheet: the upper half spoke with certainty; the final lines admitted that no full record had yet reached the town. One bookseller objected that this would damage sales. After buying his copies, he cut the bottom off.

Later Luther disappeared on the journey home. Some said he had been abducted, others that he was dead. Paul and the students worked through the night on a memorial. They had just finished setting it when someone rushed in to stop them. No reliable news had come. The paper had been cut, the ink mixed. These grieving, angry people found themselves quarrelling over who would bear the loss.

The box could not resolve it either. It kept reminding them that several letters repeated the same rumour; they did not amount to several eyewitness accounts. Only when Paul pointed out the identical spelling errors did the master agree to leave the type standing without printing. Three days later another letter suggested Luther might be safe. Everyone was relieved. Nobody undertook to pay for those three days' work.

Gradually the town learned that Luther had not been killed. Those who knew where he was would not say. Students argued as usual. Sermons carried more urgent demands; people seemed to stop heeding some of the old rules overnight. A cleric Paul knew left his monastery and stood

in the street with nothing but the books he had brought. At last he could live freely, he said. On the third day he began looking for someone willing to employ him.

Grete paid him a small sum to write several letters demanding overdue payments. He wrote well, though he always ended with a long blessing that crowded the sum owed out of sight. She asked him to rewrite them. Embarrassed, he nevertheless did.

That summer Paul and Grete married. They could afford no proper feast. His mother brought cloth from her chest and said it would be useful now. Their new home was in the street behind the print shop. The leak in the roof remained unrepaired. A little table by the wall was set aside for the box and the accounts.

Paul often rose in the night to check the door bar. He could not have said whether he feared officers or someone seizing the chance to steal the machine. At first Grete told him to lie down. Later she put a pair of shoes beside the bed so that, if something did happen, they would not have to grope for them.

In autumn, word at last came through someone they trusted. Luther was hidden at the Wartburg. He needed books, and people to find out how words were used in ordinary life. Could the thing that read be sent to him too?

Matthias refused. His agreed period of use was not over, and the debt was not paid. Paul went to him three times. In the end he pledged another six months' wages towards the debt. Grete listened, took out paper, and did the sums again. They had been charged twice for two evenings. The next day she went to have the account corrected herself.

The crate was loaded onto a flour wagon. Paul nailed down the lid and listed the books travelling with it. He remembered the receipt he had signed in France two years before: anyone who knew its origins should write.

They still did not know where it had come from. But another sheet was already needed to record the places it had been.

Chapter Four A Language for Everyone

The room in the castle was cold. Luther had changed his clothes and grown a beard. Sometimes he paused before his reflection, needing a moment to recognise himself. Greek and Latin books lay on his desk. Beyond the window the wooded hills lost more leaves each day.

Jolted on the road, the box could produce only broken sounds for the first two days. Paul's instructions lay among the books: if the brass strip underneath shifted, press it firmly to the left; do not use nails. Luther had a small knife brought and struggled with the drawing until he silenced it altogether. The steward took away the knife and found a man who repaired locks. At last a thin sliver of wood wedged into the loose joint put it right.

By the time it spoke again, Luther had finished translating the passage.

“Check this,” he said.

A word could be found for each word of the original, but the German was like an overloaded cart unable to turn a street corner. Luther walked about the room, saying the sentences to himself. A sentence that could not be understood at one hearing would not satisfy him, even if every word had a source.

“Whom are you speaking to?” the box asked.

He returned to the desk and listed several kinds of readers in a corner: mothers, children, people in the street. They still seemed vague. He remembered the voice of a woman in Wittenberg bargaining over vegetables. He said the sentence again and replaced its stiffest word.

He was not the first to translate the Bible into German. Older versions stood on his shelves for consultation. Some said that, since the work had already been done, to do it again was merely to compete for fame. He read a few pages of an older translation, thought of Paul stumbling over a letter, and kept writing. Putting a book into German was one thing; making it understood in the language people used every day still required a great deal of work.

In some places the machine was useful. Similar expressions could be set side by side, repeated quotations quickly found. Elsewhere that convenience was dangerous. It was always ready to smooth over a gap in meaning, completing a sentence where the original had left something unsaid. Luther deleted the added words and required it to distinguish what was certain from what was not. The following night it made the same mistake in another passage.

Angrily, Luther struck the table. Candle wax spilled onto the paper. As he blotted it with a discarded draft, he noticed that one of his own translations was too smooth. It had passed over a hesitation in the original.

He kept both versions and wrote to someone who knew Greek. Once the letter had gone, he could only wait. The box could list many possibilities in one evening. It could not shorten a snowy mountain road by half a day.

In Wittenberg, Paul collected expressions for him. When Grete heard something useful in the market, she dictated it on coming home. Sometimes they argued over whether a word was used that way outside Saxony too. A student laughed at them for bringing women's market talk into a

discussion of scripture. Grete asked whom he meant to sell the translation to. Everyone, he said. She repeated her expression and told him to write down all of it.

Other letters were not about words. The old order in the monasteries was weakening. Some in the town demanded immediate changes to worship; others said the changes were too fast. The cleric who had left the monastery still could not earn enough. People once dependent on church charity came to the print shop to ask what new relief would be available. Paul sent these reports as well. Luther reached the end and turned back, as though somewhere among those small troubles he might find a sentence that would order them to stop.

He found none.

One night the box drew a connection between a letter and a passage in the translation. If everyone was to read for themselves, readers would inevitably understand things differently. Which differences, it asked, could be allowed?

Luther answered quickly. The box applied his answer to several examples. One was a view he favoured; another approached the position of people in the town he opposed. He told it to distinguish them again. They talked until the wick burned low. At last he put the letter in a drawer and returned to his translation.

Events in Wittenberg did not wait for him to settle the question. People acted for themselves. Disputes entered the chapels and their homes. Paul's mother would not abruptly abandon rituals she had known for years. A neighbour called her timid. She listened without replying. When the visitor had gone, she asked her son whom she was supposed to pray to now.

Paul thought for a long time, then told her to pray as she always had. His mother nodded, but her face did not quite ease. He realised that he too had told her what she should do, as though receiving a few pages of a new translation entitled him to decide what someone else should believe.

In the spring of 1522, Luther left his refuge and returned to Wittenberg. His sermons tried to slow the rapid changes. Yet much that had happened could not be undone by a series of sermons. Broken things needed repair; people who had lost their support needed work. Those who believed they acted for the truth did not welcome being told they had gone too far.

The box came back with him. Paul opened the crate, saw fresh scratches, and checked the brass strip underneath before anything else. Luther gave him a stack of manuscript pages and asked him to set a trial passage. The book still needed repeated checking. Melanchthon and the other collaborators each had objections. Some words survived; some sentences were translated afresh. A page would scarcely be set before another dozen changes arrived. When Paul complained, Luther stood beside him waiting, with little else he could do.

Grete marked the sentences that did not read easily. She knew no Greek and could not settle questions about the original meaning for the scholars. She could tell them only where the language failed to sound like ordinary speech. At first nobody treated her marks as a formal contribution. After one corrector twice adopted changes she had suggested, he asked her name.

She told him. He nodded, and in his letter wrote “a local woman”.

When Grete saw it, she did not quarrel. At home, though, she put her own copy of the manuscript in a separate wrapper. Paul asked why she was keeping it. So that she herself would

not forget which sentence she had changed, she said. He wanted to have her name added to the letter. She stopped him: work at the shop would begin early tomorrow; first they should sleep.

One evening shortly before printing began, a bookbinder and his wife brought folded gatherings. He had heard this book would be available to buy. How thick would it be, and what would it cost? Paul gave him the provisional price. The craftsman's hand moved in his pocket without coming out.

“Could I buy part of it first?”

“You can buy unbound sheets.”

“I don't mean that. I mean only as much as I can read, each time.”

The man went with Paul to count the gatherings. His wife waited until the others at the table had moved away, then reached towards the German proof. Paste remained beneath her nails. Afraid of dirtying it, she touched only the edge with a fingertip.

“Am I allowed to read this too?” she asked.

Luther turned the sheet towards her.

“Can you read?”

“A little. I've kept accounts for people.”

“Then begin with this line.”

She did not move. To her, these were words only a priest was permitted to explain. If she read them wrongly, whose sin would it be? She gave a little laugh, as though apologising in advance for the question.

Luther put down the pen he had just picked up.

“All are equal before God. Your baptism is no less than mine.”

She looked at him. Slowly she said, “But you are a doctor.”

“Then I ought to help you understand.”

He read her the first sentence. She listened, then read it herself, stopping at a long word. Luther taught it syllable by syllable. The second time she got it right, pressing a faint mark into the paper beneath it with her nail.

Someone beside the table could not resist reading on for her. She laid her palm on the sheet.

“Wait. I'm not there yet.”

The reader stopped. Luther moved the stool to one side, making room for her in the light beneath the window.

She read very slowly and made two more mistakes. She asked Luther about a sentence she did not understand, and still frowned after his answer. He explained it another way. This time she nodded and returned to the line.

Before leaving, she asked whether she could take the discarded proof home. Her daughter wanted to learn to read too. The back had been spoiled in printing, and the front bore several changes. Luther supplied the missing words with his pen and waited for the ink to dry before giving it to her.

“Read it together in the evening. Leave the parts you're uncertain of for now.”

The bookbinder had finished counting the gatherings with Paul and was waiting at the door. She tucked the single sheet into her apron separately from the rest. At the door she turned back to ask where to divide the long word.

This time Luther did not read it for her first. He pointed with the nib and waited for her to say it herself.

The first printed sheets were still wet and had to be spread out to dry. There were not enough racks. Grete gave over the line where she dried her washing. That evening her clothes lay in a heap at the foot of the bed, while newly printed pages filled half the room. She ducked beneath them to open the window without damaging a single sheet.

Chapter Five The September Buyers

When the September translation went on sale, the first buyers were not the poor people with handfuls of coins Paul had imagined. Officials came, prosperous craftsmen, stewards buying for their employers. Some ordered several copies and asked for handsome bindings. When word of the price reached the back street, Grete heard a woman say that much money would keep her family for a while.

In the end they pooled their money for a set of loose sheets. Four households paid a share; a fifth provided lamp oil. Whoever hosted the reading would watch the children and keep their hands off the ink. Paul trimmed the sheets and borrowed the shop's needles and thread to bind them. For the cover he used old leather Matthias had thought the wrong colour for a gift.

Eleven people sat in the room on the first evening. Only three could read fluently. They took turns; the others listened. Paul's mother, whose sight was poor, sat nearest the book but always let someone else read in her place. Grete brought her a chair with a back without asking again whether she wanted one.

Before they had read a page, the dyer's wife interrupted. Her question had never occurred to Paul. Another reader thought it not worth the delay. They argued, the open book between them, and nobody continued. Grete wrote the question on a scrap left from trimming and slipped it between the pages. They agreed to return to it when someone could answer. Little was read that night, but all the lamp oil was used.

Seeing Paul drowsy the next day, Matthias asked who paid for his evening work. The question was not merely a taunt. The shop was desperately busy, and one wrongly set line meant work for several people. Paul promised to take care. That afternoon he still put two similar sorts into the wrong compartments. His master made him sort the whole section again. Grete had to lead the evening's reading.

She read slowly and began again when she made a mistake. One woman went home halfway through; her husband disliked her staying out so late. Another missed the second meeting because her employer had given her extra work. Equal shares in the book had not bought each household equal leisure.

Paul borrowed the box to read the difficult passages aloud. The children were excited at first, and gradually the adults gathered round too. The machine neither paused for breath nor hesitated. It sounded better than Grete. One woman, absorbed in listening, used its words to pass a harsh judgement on her own marriage. Grete sensed something wrong and checked the page. The box admitted that it had mixed in an explanation of another passage.

The room fell quiet. The frightened woman drew her shawl closer and asked which part she could still trust.

That evening, for the first time, Grete would not let it speak uninterrupted. After each passage someone checked its words against the page, then asked whether the listeners had questions. This was slower. The children began yawning. But the following week the woman returned, bringing two words she had written down herself.

Within months, the book had travelled far beyond Wittenberg. Some reprints were cheaper; some more beautifully printed than the original. Matthias began receiving unsold stock back. Someone in the market, he was told, sold the same text for less. Furious, he wanted to bring a suit. Then he discovered that the seller was based outside the town. Finding whom to petition and what loss to claim would cost more money.

On hearing this, Luther first asked whether the reprint had corrupted the text. The master first asked how he was to pay his workers. They were talking about different things, and neither could dismiss the other's concern. Later a small print shop wrote asking for corrections. It would not pay a high price for the original manuscript but was willing to make the same amendments. Matthias refused. Paul showed the letter to Luther, who insisted that the list of known errors be sent.

“And who pays for the paper?” Matthias asked.

Luther felt in his pocket. He had not brought enough and sent someone home for more. Matthias said nothing further, but the next day he altered the shop's terms of credit for paper.

Paul had once thought that knowing the right thing to do left only the question of whether one would do it. He was becoming familiar with another situation: everyone knew the letter should be sent, yet it lay on the table for two days while someone found the money.

The readers in the back street cared little about these distant disputes. They wanted their copy to hold together, and to be able to find their earlier questions when they borrowed it again. Grete found an old ledger. She left the settled debts in the first half and used the second to record questions page by page. Where an answer was uncertain, she left a blank. If a later explanation differed, she did not erase the earlier one.

At first people mocked this. They could not understand one book and were making another full of questions. Grete admitted she did not understand and kept writing. It was then that Riedel came. He had once taught Latin and now wrote letters for merchants. Borrowing the box to check an inventory, he looked through her notebook and asked whether he might help with the reading once a week.

“For how much?” she asked.

He named a very small sum. Seeing her hesitate, he explained that he too had rent to pay. This reassured her. It was easier to imagine working for years with a man who needed rent money than with someone who promised to work for nothing for ever.

In spring the readings were divided between two evenings. Beginners could learn their letters at the later session without always holding back those who could already read. Anyone could still bring questions to the earlier one. For the first time Paul's mother wrote her own name on paper. It was crooked. She folded the sheet into her pocket. This did not make her good at arguing. Asked for her opinion, she still said she did not understand. But the next time someone offered to explain the old indulgence certificate, she took it out and asked them to read the words actually on it.

The print shop held no special celebration. That very day a batch of paper was ruined by damp, at a loss greater than the previous month's profit. Paul and Grete moved the books onto the bed and sat on the floor while the roof was repaired. The box suggested ways of preserving wet

paper. One required equipment they did not have; another the master rejected outright. In the end they followed his old method, separating the sheets, pressing them flat, and letting them dry slowly.

Among the damaged pages Paul found a reprinted pamphlet. Neither author nor printer was named on the last page. The paper was poor, but the type was clear. He meant to use it beneath a table leg. Grete kept it: a child was due to join them the following week and needed a copy.

“We know there are errors in that edition,” Paul said.

“Then help him correct them.”

By the last of the daylight they copied the corrections they had into the margins. The child came to collect the book the next day, bringing a small bundle of firewood in thanks. Grete accepted it. During the reading that evening, the fire lasted a little longer than usual.

Chapter Six Readers Must Eat

On the day Lena was born, Paul was setting a text urging the establishment of schools. It was 1524. Luther was calling on the councillors of German towns to provide schools for children. At the passage about girls, Paul read a second time. His master told him to hurry. Grete was waiting at home.

The child was smaller than he had expected, her cry very loud. Paul washed his hands three times, but could still see ink beneath his nails and was afraid to take her. Too tired to laugh, Grete told him to support the back of his daughter's head and stop treating her like damp paper that might crumble.

After holding her for a while, Paul began describing what he was printing. At the mention of schools, Grete asked with her eyes shut, “Who will feed the teacher?”

He had not thought of that.

Soon Riedel brought him a petition for a literacy class. Old sources of support had been cut off; some existing schools could no longer continue. Many people had designs on monastic lands and buildings. Fewer were willing to set aside a regular income from them for teaching. Riedel hoped to borrow an unused room, taking boys and, at another hour, girls. Reading, writing, and arithmetic could begin there.

His repeated visits won much approval. When it came to writing down a sum, everyone urged him to keep within his means. Matthias would contribute, provided his own employees had priority and more time went to trading accounts, weights and measures, and letters. A guild leader who offered money opposed admitting girls. His contribution ought to be spent on something “useful”.

Holding the baby in the doorway, Grete produced the contract she had corrected for Matthias some years before. The guild leader could not follow the detailed accounts in it and passed it to a companion. She asked no further question. When they had finished, she took the paper back.

The school that finally opened was smaller than Riedel had hoped. They could afford only a room that let in winter draughts. The girls came later, after lamps were lit, making it difficult for those who lived far away. Some families would send their children but could not spare their labour. Riedel moved the lessons later, then ran up against employers' rules forbidding apprentices to go out at night.

He had imagined that all he needed was willing pupils. A month after opening, his records were full of reasons children could not come: younger siblings to mind, goods to deliver, a sick mother, broken shoes. An eleven-year-old boy missed three lessons in succession. Riedel went to reproach him and found that his father had injured a hand. The boy was carrying the family's water.

“Will reading earn me money?” he asked.

Riedel crouched down. He did not make a promise at once. Reading would open more kinds of work to him, he said, but someone would still have to carry tomorrow's water. He helped with

a bucket. The next day he shortened the lessons and went in search of someone willing to pay for lamp oil.

One evening each week the box was brought to the school. It could repeat a word patiently, freeing Riedel to teach another group. Watching people of different ages around the table, Paul began to see what could be done with the time a machine saved. He also saw a child invent a meaning for a word. The box accepted it and elaborated, while everyone round it collapsed with laughter.

The teacher did not punish the child. He required the errors to be kept and checked one by one the next day. At first the boy was proud of having beaten the machine. Faced with explaining to everyone what was wrong, he began to evade the task. Riedel made him stay and finish it.

Gradually the school attracted craftsmen. A young carpenter brought a waterwheel sketch and asked how to make it run with less effort. The box helped calculate the gear teeth and suggested a modification from an old book. The drawing looked good. The wooden gear they built jammed under load. The young man returned to complain, his sleeves dark with water. He had wasted timber and been scolded by his master.

Paul went with him to the workshop. The record he brought back was less tidy than the drawing: knots in the timber, an axle hole drilled off-centre, water flowing sometimes strongly, sometimes weakly. Only with those details could the box revise its answer. In the end the carpenter kept one small improvement and restored the rest to its old design. He hung the broken gear on the wall. He would no longer let anyone claim that putting a new method into practice was simply a matter of following a drawing.

Matthias was interested and offered to buy the improved design. The young man hesitated. His master said that, since he had provided the workshop and timber, the drawing belonged to the workshop. Their quarrel reached the school. Riedel had no authority to decide it, and neither did the box. Grete asked them to list what each had contributed. They could not even agree on who had first proposed improving the wheel.

The agreement they finally signed was clumsy. The workshop could use the design; the young man would keep a copy; Matthias paid for the right to resell it elsewhere. The school retained only an anonymous account of the failed trial. Nobody made the fortune they had imagined. Yet after that more questions arrived: why dyes faded, how bridge planks could resist bending, how to convert the distances used along trade routes.

Riedel separated the questions, so that a child just learning letters would not lose a seat to adults arguing over gears. Grete hung a curtain across half the room and told them Thursdays belonged to the children. Matthias thought this wasteful. Thursdays brought the most craftsmen. Craftsmen could change their day, she said. Some of the children's employers would release them on no other.

This dispute proved harder to settle than the translation of a word. Matthias supplied the money. He had every reason to want it used effectively. In the end Riedel accepted lower pay to preserve Thursdays. When Paul learned of it, he added a line to his own accounts, intending to make up the difference later. There was no spare money that month. The entry stayed blank.

In late spring Grete's brother Jakob came to town from the country. He brought a sack of beans and a roughly carved wooden spoon for his niece. After supper he took several sheets from

inside his coat and asked his sister to read them. In different hands they set out matters the villagers had discussed: rents and taxes, woodland, and fishing rights.

Halfway through, the child woke. Jakob took her and listened as his sister continued. If they too could read the Gospel, he asked, why were some people still born as the property of others? Why could he no longer gather firewood in woods his father had walked through?

Paul said translating a book could not settle that. Jakob looked at him without anger.

“I know. I haven't come for a book. I want someone to help us say clearly what we mean.”

The box stood at the table. Paul put the loose sheets in order and held them one at a time before the slot. It could arrange confused items neatly, but could not decide the villagers' demands for them. Jakob interrupted several times. A ditch that had disappeared into a general statement must be put back. That ditch had a name. The village depended on it.

Grete took her daughter back and listened to her brother's corrections. He seldom spoke at such length before educated men. Even at the end he feared he had misrepresented his own meaning and looked at his sister. She told him to go on.

Chapter Seven The Twelve Articles

In spring a printed copy of the Twelve Articles reached Wittenberg. Paul opened the parcel and saw familiar words: the Gospel, common rights, grounds for correcting an error. Peasants in different places had different grievances. No single text directed them all. But this printed document gave scattered anger a form that could be carried.

Jakob read several articles and wrote that some concerned matters his own village had disputed for years. He had not needed the pamphlet to feel wronged, nor the new translation to know hunger. Before, when someone told him he must not question, he had often had no reply. Now he wanted a common ground on which they could argue.

He enclosed a list with his letter: which wood had been closed, whose animals had been confiscated. Paul gave it to the box. It quickly composed an impassioned appeal. When it had finished, Grete asked where the actual people had gone.

“They have been grouped under matters of the same kind.”

She told it to begin again. Not one of the names her brother had laboured to write could be left out. Paul added them at the edge, discovering that the tidy text had left too little room.

An order came to the print shop to print the demands. The master looked at the price and said nothing for a long time. Matthias opposed accepting it. He had sold books criticising indulgences, but would not print something likely to anger lords and local officials. Peasants' payments were uncertain, transport dangerous, and nobody would stand surety if something went wrong.

Paul pointed to the bundles of polemical tracts against the wall. Those offended people too. At least, Matthias answered, they had protectors who could do something.

Everyone knew it was an ugly thing to say. Everyone knew it explained many entries in the accounts. The master agreed to let them set the text and take a single proof for copying. He would not print a run. Paul did the work after his shift. Next day a student tucked the proof into his coat to take to another town. No bundles would leave this shop, but the words would still find a way out.

When Luther's first admonitions arrived, Grete asked Riedel to read them to the group. They reproached the lords and warned the peasants. Jakob was absent. A farmer in town to buy salt listened and said some of it was right. But what was he to do if, when he returned, he still had those rents to pay?

Nobody in the room could suggest anything he could put into practice the next day. A student spoke of patience. The farmer looked at the student's unworn shoes and silently tied his sack of salt. After he left, Riedel did not turn the page for a long time.

The reports that followed grew more confused. Monasteries were entered by force, nobles killed, manor houses burned. There were negotiations too, covenants against looting, villages afraid of both sides that wanted only to keep the armies away. Each letter described the small part its writer had seen. Printed as news, it could easily pass for an account of the whole region.

Matthias would print news of rebellion because it sold. He wanted a more alarming woodcut on the cover. The cutter said he had seen none of it. Neither had the buyers, Matthias said. Holding the block, Paul remembered the names and woodland in his letters. Before printing he finally removed an unverified sentence. He was sharply reprimanded. The sheet still went to press. He could not stop its other exaggerations.

Then Luther wrote in harsh terms supporting the suppression of the revolt. When Paul first read the text, he could not believe the signature. He took it to another student, who confirmed its authenticity and began defending its words. Paul left before he had finished and brought it home to the table.

Grete did not ask the box to read it. She read very slowly herself, twice going over one passage. Her daughter stirred in the cradle. She went to pat her, then returned to the page. At the end she asked Paul whether this too was something people would read once they had been taught their letters.

Paul said yes.

That night she put the book in the cupboard without tearing it. Had she torn it, Paul would have found it easier to comfort her. The next day she simply changed the school lesson to the names of several villages and asked the children to identify where the letters had come from.

Still no word came from Jakob. Grete asked every merchant arriving from the south. Some said he was alive, others that he was dead. From several letters the box worked out a route he might have taken. Paul followed it and waited uselessly at a posting station for two days. Seeing him come home alone, Grete gripped the back of a chair. Neither asked the machine again which route was more likely.

They eventually found Jakob in a barn temporarily sheltering the wounded. He was alive, with a leg injury and a face grown almost unrecognisably thin. Even after the defeat, the fines and searches went on. Villagers dared not shelter him openly. Grete went to change his dressing. The old cloth clung to the flesh. She fetched warm water before trying to lift it.

Jakob asked whether they still had the petition at home. Paul said they did.

“Don't turn it into a story about our asking for too much,” he said.

Grete bent her head, wetted the cloth, and loosened it a little at a time. She promised.

Bringing the injured man into town led to another dispute. The school's supporters feared being implicated. Some parents kept their children away. Riedel arranged for him to stay elsewhere. Grete opposed him, and they fell out. In the end an old worker from the print shop made room in his loft for a small payment towards fuel. He declared that he endorsed nobody's cause. He was simply taking in an injured relative.

That declaration could not protect everyone. It did give some people a reason for helping that they felt able to defend. Several households from the reading group took turns bringing broth. The student who had urged the farmer to be patient came once, left medicine, and went away. On his second visit he stayed to change the water.

Jakob recovered slowly. He no longer liked loud speeches, but often asked his sister to read the specific demands. Visiting villagers first asked about prisoners, harvests, and fines, then spoke of faith. Paul took notes. After defeat, the same words sounded utterly different. Some still

wanted to dispute them; others hoped never to mention them again. Some thought they themselves had not been resolute enough.

No explanation could put them back together as they had been.

A few months later Grete took the list of names to Luther. She came with no elaborate argument, asking only whether he had read what had happened to these people. He said he knew many dreadful things had happened. The breakdown of order would hurt still more innocent people, he said, and explained why he had reached his judgement.

Grete listened with her hand on the paper. When he finished, she left the list on his desk. Several names still needed checking, she said. She hoped he would read letters from more than his own side.

Paul waited outside. When she came out she did not say whether the conversation had done any good. She asked him to support her for a little way. At home the child needed care, and Jakob's dressing was due to be changed. Passing the print shop, they could still hear the press, stroke after stroke, clearly through the thick door.

Chapter Eight The Type Kept Moving

Katharina's first proper conversation with Grete concerned the loan of a large cooking pot. She had been out of the convent for some time, living under other people's roofs while learning to arrange her own life, and to say no when others chose a husband for her. In 1525 she married Luther. Suddenly many people cared about her every movement. Few asked how the meals for a large household were actually made.

Grete brought the pot and found a yard full of things that needed sorting. Sleeves rolled up, Katharina told the men moving them what could still be used and what could burn, while also attending to visitors. She accepted the pot and asked whether Grete wanted rent for it or a favour in return.

“I want it back in one piece,” Grete said. “Don't burn through it.”

Katharina laughed. That was how the two women came to know each other.

The disputes outside had not stopped for the marriage. Some attacked Luther for it. Others printed cheerful pamphlets, trying to make a household prove a point. Katharina read one and found her name among sentences that were far too pretty. She asked the author why he had not visited before writing. He said the words were a blessing. She asked him first to carry the bundle of wood at his feet into the kitchen.

Grete sometimes helped with the accounts. Beer, grain, lodgers' board, money owed by students: each presented difficulties. Some distinguished visitors took their stay for granted and forgot the cost of meals when they left. Luther told his wife not to quibble. She pushed the household accounts towards him and asked him to point out what they need not buy next week.

Brought in to help, the box quickly found two accounting errors. Katharina was pleased. Then she asked when the debtors would pay. Its estimate, based on past records, was soon disproved by a lodger who fell six months behind. She did not blame the machine, but wrote beside the date: Cannot buy grain on this assumption.

Jakob, still in the print shop's loft, could now get downstairs holding the rail. He could not return to his old work. He offered to make repairs for the school in exchange for help with his fuel costs. Riedel asked for a lower bench. Jakob built it to the measurements, but it rocked when the children sat on it. The floor was uneven; the legs needed cutting again. He was furious. Later he admitted he was angry at himself for no longer being able to make even a bench properly.

Grete put the old wooden spoon into his hand and asked him to smooth its burrs for Lena. He had brought it the evening he came with the petition. Jakob sat feeling its handle for a long time. The next day he began repairing a second bench. Suffering had not made him gentle. His temper remained bad. At the sight of passing soldiers his hands would suddenly stop. If anyone spoke of the dead as a mob of fools, he always quarrelled.

The school gradually recovered, smaller than before. Donations stopped twice. Some parents returned; others never did. Riedel put a notice by the door listing only what was taught, not who stood with whom. The students were disappointed. He had become timid, they said. He admitted

he feared the school's closure. If it shut, they would have other places to argue. Some children would never come again.

When Paul spoke to him about this, Riedel was mending his coat. His weekly pay had fallen again, but clothing could not wait for quieter times. Could the box answer three readers at once? Paul tried it. With too many questions the machine confused the speakers. An answer about timber for Jakob appeared beside a woman's prescription. Next day they made a queue. Each questioner must give a name. Anyone hurrying to interrupt had to wait again.

A travelling bookseller called the rules tiresome. Why not buy a volume of ready-made explanations and save the trouble? What if its explanations were wrong, Grete asked. They could wait for a reprint, he said. She remembered Jakob in the barn. Some errors travelled far faster than any corrected book.

In 1527 plague approached Wittenberg, and many people left. Teaching, bookselling, and disputation were all disrupted. Those who could afford to move went first. Those who remained had different reasons for being unable to leave. Some stayed to nurse others; some simply had nowhere to go. Luther and Katharina took the sick into their home. Arrangements once fiercely disputed had to yield to beds, water, and clean cloth.

Riedel stopped lessons. Grete brought old sheets. The machine could repeat instructions for nursing and record when someone had drunk. It could not carry water to a bedside. Asked whether a patient would live, it gave a long answer. Katharina stopped it and asked it to repeat the number of cloths needing washing. She did not need a voice as composed as a physician's to make the waiting seem easier.

A poor woman dying there could not stop worrying about her son. She asked again and again whether he had somewhere to go. Paul approached three households. None dared promise at once. At last Riedel said the child could stay with him for now. How long could be settled later. His room was already small. To fit another bed he had to move the bookshelves beside the window. Winter damp mildewed several books.

When lessons resumed, the child attended. He learned more slowly than the others and kept forgetting the same letter. A visitor willing to donate watched for a while and said more money should go to clever children, so it would not be wasted. Riedel politely saw him out. Returning, he found Katharina beside the boy, asking him to count clean cups and give one to each person.

He miscounted once and began again. When the last cup stood safely in place, she took her own and said that was enough.

Grete later asked what they would do if this cost them donations. They would think again, Katharina said. If every shortage were first met at the expense of the least able to resist, the household accounts would look excellent, but the house would be impossible to live in. Then she bent over the figures again. She did not promise she would always manage it.

Paul's mother died that winter. She did not live to see Lena read. In her last days she heard her granddaughter call to her. Among her things Paul found the old certificate, Luther's letter, and the paper on which she had practised her name. All three were wrapped in the same cloth.

He meant to keep only the later two. Grete told him to keep all of them. His mother's life was more than the small part that remained after they had corrected it for her.

Early in spring Matthias brought cheap paper, on condition that manuscripts he chose took priority. Without more paper the shop would have to work two fewer days each week. Paul signed the receipt, put his mother's bundle back in the cupboard, and went downstairs to carry paper. Jakob helped too. Unable to move easily, he stood by the door passing the bundles in one at a time.

As the press bar began moving again, Grete was teaching Lena to distinguish the marks on the type case. The child pointed to an empty compartment and asked why it had no letters. Grete said they had been taken out and not yet returned. She led her daughter to the forme and showed her where used type belonged when the work was done.

Chapter Nine Whose Interpretation

Two kinds of news came back from Marburg. One said agreement had been reached on many questions; the other that serious differences remained at the end. Paul spread the reports out. Neither had lied. They had simply put different things first.

At the meeting in 1529, agreement had been reached on most of the fifteen articles. The division over the Lord's Supper remained. Listening to accounts of Luther's dispute with Zwingli, Paul struggled to understand why a word could not be put another way. Riedel reminded him that, for those disputing it, this was more than a question of wording. To yield would mean surrendering what they believed to be true.

“What are the people here to do?” Grete asked.

Nobody answered at once. Several readers had already been influenced by sermons from elsewhere. They still paid for lamp oil and helped newcomers learn their letters. Only yesterday Paul had carried timber with one of them. He had not imagined that today they must first establish how each understood the Lord's Supper before sitting at the same table.

Matthias brought new conditions. His trade routes crossed towns that accepted different books. To reduce losses, the explanations used in the school should first be read by an approved person. The teacher he recommended would charge nothing. He asked only that no unapproved interpretation be circulated.

“I haven't told you not to read,” Matthias said to Grete. “I only want to know what my money is buying.”

Riedel understood the demand. Others supported the school, and the town could hardly be indifferent to what was taught. He proposed separating children's literacy lessons from adult disputation, leaving some days for those who still had questions. The school's backers were willing to discuss the hours, but would not keep the words “still had questions”. To them, that sounded like permission to refuse instruction indefinitely.

Listening, Paul remembered his mother asking whom she should pray to. Such uncertainty was not impressive, but it was real. In print people preferred clear conclusions: short, memorable, and convenient for checking whether others obeyed.

To try the new rules, Matthias asked the box to turn several prescribed books into questions and answers. It quickly produced a concise set of explanations. Even Paul found it useful. Riedel taught from it for a month. The children recited more fluently than before, and the inspectors were pleased. Once, though, Lena asked him why. Her question was not one of those in the book, but out of habit he pointed to the next item.

By the time he realised, she was repeating the standard answer. He stopped her and asked her to repeat her own question. Lena thought for a long time. She had forgotten it.

The adults' disputes were harder to manage. A young man asked the box to put his own interpretation into words. The orderly sentences looked as though they had come from someone with years of training. Another brought a prepared rebuttal. They stopped listening and took

turns reading aloud. Grete collected both sheets and asked each to explain in his own words what he disagreed with.

One could not. Ashamed and angry, he accused her of having no right to conduct such a discussion. She agreed she had no authority to settle doctrine. But she paid part of the rent for this room, and should have a say in who was allowed to speak. The young man left and spoke elsewhere of her arrogance. A donation disappeared.

This time Riedel did not take her side. He had rent to pay. More children had joined the school. They could not afford to lose another source of income with every dispute. They argued late into the night. Unable to calm them, Paul took Lena next door. In the end Grete agreed to announce the subject and limits of a discussion beforehand. Riedel agreed that disagreement alone could not justify striking someone off the roll.

They signed two respectable-sounding lines that proved difficult at every turn. What if an objection concerned the town's orders? What if someone repeatedly interrupted? What if the largest contributor wanted an exception? Each time, they added a clause on the back. Soon it was full.

Riedel complained that he was becoming a clerk. Grete said a clerk at least had to account for figures that had been left out. He was angry for a while and taught as usual the next day.

That year Paul came to know a scribe named Isaac through his trading contacts. Isaac worked in another town and knew Hebrew, which Paul did not. At first he deciphered only a page of quotations; later the letters multiplied. Isaac asked directly why, if he was to take responsibility for corrections, explanations forwarded to others said only “someone has pointed out” and omitted his name.

Paul showed Matthias the letter. The merchant frowned. Different buyers had different objections. Printing that name might make a whole batch difficult to sell. Paul asked Isaac whether he would use another name. He refused. He did not want his address made public indiscriminately. Nor did he want to contribute his learning and then lose even the right to decide how it was credited.

They finally kept his name in the corrected copy they used themselves and asked his consent before sending it anywhere dangerous. This was far more trouble than simply erasing it and cost several rounds of postage. Matthias asked whether they would now have to consult every person who corrected a word. Paul had no answer. He did not know how consistently they could manage it either.

The box preserved these exchanges. Later, if someone asked only for approved explanations, it supplied only those. Unless listeners knew to ask another way, they could easily think no other views had ever existed. Paul added a note at the beginning of each record, naming the books used and those excluded.

Riedel considered it for a long time. His lessons should begin with such a note too, he said. He would teach the catechism approved by the town, but would not pretend the children's other questions had never been asked. This did not free them from inspection. It merely left more papers to explain whenever an inspector came.

Luther once came to look at the records. In a margin he wrote an objection in heavy strokes. Grete had expected him to demand that the page be torn out. He did not. He told her only where he thought it wrong. She felt relieved, then heard him add that some teachings could not be allowed to spread.

She understood that he saw no contradiction. Nor could a single sentence of hers secure a hearing for every question. After he left, she put the annotated page back, leaving room for both hands.

At dusk Lena crouched on the threshold practising her letters. She wrote “ask” larger than the other words, thought it untidy, and rubbed away a corner with her sleeve. Grete went over. She did not correct it for her, but handed her a small clean cloth.

Chapter Ten The Second Schoolroom

By 1534 Lena was ten. She could read many sentences her mother could not have read as a child, and sort proofs for her father. When the new translation of the complete German Bible was printed, an unbound copy stayed at home. She needed both hands to carry its heavy stack of pages securely.

Years of work had gone into it. People had checked the Hebrew, the Greek, and different interpretations. Paul had watched scholars dispute a word, then return the next day to overturn yesterday's conclusion. He no longer imagined translation as one person's work in a quiet room. A messenger, someone who lent a book, a person who knew livestock or farming tools might supply just what a particular page needed. Their names seldom travelled with the book.

Isaac sent a long letter correcting a statement about a plant. Comparing it with older records, the box pointed out that a countryman had raised a similar question years before. His rough wording had led to its being filed as irrelevant. Paul found the sheet. On its reverse was a half-settled timber account.

Sitting beside him, Jakob recognised the man. He had moved away; finding an address would take work. Paul wrote anyway and asked an acquaintance to forward the letter. Two months later a brief answer asked whether, if the new book was corrected, they could send him a copy to see. He did not congratulate himself. He was still concerned with the small point he had never managed to explain, and glad that someone would at last hear him out.

Before the correction could enter the next printing, someone who knew the original language still had to approve it. Grete did not demand that the countryman's suggestion be accepted as true merely because he had made it. She asked Riedel to keep the whole account: how a comment moved from “irrelevant” to worth examining, and how examination might still disprove it.

The first school had acquired rules. Strong readers could proceed to more advanced lessons. At a certain age most girls were encouraged to go home and learn other things. Riedel neither fully agreed nor publicly opposed it. He knew that a breach with the funders would leave him able to promise the children even less.

The first time Lena was kept out of a lesson, she was told the subject was “not suitable for girls”. At home she did not cry. She asked her mother whether the books really contained something she must not see. Grete opened them. She could not understand them either, but remembered a time when the school had been less certain about who belonged in which room.

She asked Katharina for the loan of a temporarily unused room, proposing lessons on two more evenings. She did not intend to fight the whole school. The classes would take people who knew the basics and wanted to continue, boys and girls alike. Katharina agreed to try it for a winter. She made clear that the room would later be needed for something else. They could not occupy it free for ever.

Grete offered the money she had saved from years of mending and keeping accounts. Paul contributed too. They invited Riedel, who took a long time to reply. A few days later he appeared

at the door with a bundle of books. He could teach, he said, but would not conceal it from the first school. He had told them already and might lose some of his pay.

There was no grand opening. Five people came the first evening, three the third. A strong, quick-witted apprentice lacked basic reading skills, and the others grew impatient at waiting. A merchant's daughter learned quickly but complained of the cold. For a time Grete feared she was spending money on a pleasing idea that would never work.

Jakob stopped up the gaps round the window. Katharina borrowed an old stove. Paul changed one of their allotted evenings with the box and paid the difference. Attendance slowly settled. Some brought money, some wood, some swept the room before class. Those who contributed less did not have to explain themselves publicly, but the accounts still had to make sense to those who paid.

At first they used the box to help read Latin. Later a carpenter brought another question. The old gear records survived, but a new apprentice doubted the measurements. He could not reproduce the thing from the dimensions on the page. They invited the original craftsman. Older now, he was offended at first. At the sight of the jammed wheel he remembered a washer he had left out of his account.

The addition took less than half a page and was more useful than a whole evening praising new inventions. Lena stitched the original record, the added explanation, and the results of a fresh trial together, leaving several blank pages. Matthias proposed making them into Questions and Answers for the Trades, to be published before the fair.

The offer tempted Grete. Riedel needed his pay; their time in the borrowed room was nearly over. Sales might carry them through the winter. But Matthias wanted the failed trials removed, leaving only reliable methods. Unresolved questions should be given definite answers too.

“Buyers pay for answers,” he said.

Lena asked whom the buyers should come to when something made from the book went wrong. Matthias laughed at how she sounded like her mother, then proposed adding a notice at the front: craftsmen proceeded at their own risk. Jakob took the old gear from the wall, put it before the bookseller, and asked him to feel the weight of timber spent on a mistake.

The pamphlet they printed was thin. It included only the few items already checked, with notes on what remained to be tested. It did not sell widely. The first edition scarcely made a profit. The following year a craftsman elsewhere sent an account of a different failure he had encountered while following its instructions. That helped them find the source of an error they had not yet resolved.

The pages began to resemble letters passed back and forth. Wherever they arrived, something might be added. A name on the cover could not vouch for all these later additions. Riedel dated each edition. Lena numbered and kept old copies even when pages were missing. Paul complained of the space they took. Then a letter about a dispute revealed that buyer and seller held different editions. The missing page was the crucial one.

Someone once asked whether the box could complete the records alone. It said it could organise what was brought to it and identify contradictions. It could not see a wheel never brought to its table, or guarantee the accuracy of a correspondent's measurements. Lena wrote

down these limits and fixed them beneath the slot. The children mostly ignored them. Regular visitors were often brought up short by those few lines.

Near the year's end the first school required Riedel to give up his outside classes. Its reasons were not groundless. His strength was limited; he was tired during the day. Several parents worried that the two schools taught differently. He did not resign at once. Grete almost quarrelled with him over it. Katharina drew her into the kitchen. First make sure he could eat next year, she said, then demand an honourable choice.

So they worked the accounts through to the following autumn. Several households would make small monthly contributions towards the shortfall. Anyone unable to continue must say so beforehand. In the end Riedel reduced his hours at the old school without leaving altogether. On the day he came to the new room he brought his own chair, discovering on the way that a leg was loose. Jakob repaired it with an old nail.

Before the lesson Lena put the chair beside the window. The teacher pointed out that, with the light behind him, the pupils could not see what he wrote. She carried it back. They moved it three times before settling on a place.

Chapter Eleven A Name on the List

It was Lena who noticed Isaac's name had vanished from a record of corrections.

She was nineteen and could handle most of her father's correspondence. When a new printing arrived, she habitually checked it against the previous edition page by page. Half a line identifying the person who had raised a question was gone, although the resulting correction remained in the text. She asked whether the type had fallen out. No, the master said. The person who delivered the manuscript had wanted it removed.

“Why?”

The master took off his spectacles and told her to ask the man who had brought it.

By 1543 people in the town had read Luther's attacks on the Jews. Challenging authority in his youth had not kept him, in old age, from using his own authority to harm others. Some disputed the harsh proposals in his writings; others endorsed them. They also served to justify existing hatred.

Lena did not want to believe the altered edition was connected with this. She went to the man responsible. He admitted the name would disturb some readers and was better omitted. Isaac's learning was still useful, but the man himself was treated as a blemish to be removed.

“If we credit him with the correction, are we to sell the book or not?” the man asked.

At home Lena laid both editions before her father. At first Paul said only that he would take it up with them. Grete asked when. Once this batch was printed, he said; otherwise all the copies already done would be lost. She looked at him without speaking. Paul turned away to find his coat.

Isaac did not live in Wittenberg. Since the stricter bans in Saxony, their contact had depended increasingly on trade across borders and trusted intermediaries. Paul knew this, yet had kept mistaking “letters still arrive” for “his life must be reasonably secure”. This time, when no answer came for a long while, he discovered he did not even know a second address where Isaac could safely stay.

Matthias seldom handled shipments himself now. Relatives had taken over most of the business. The young man responsible was more cautious than he and refused to carry the parcels. A little extra freight money could not cover the risk. Angrily, Paul demanded to know whether he approved of those views. Approval made no difference to the way officers searched, the young man replied.

That did not make his refusal harmless. It did mean anger alone could not solve Paul's problem. He finally entrusted the letter to a merchant familiar with the lesser roads, paying in advance and agreeing to make no claim if it was lost. As Grete brought out their savings, she included the money for next month's lamp oil at the school. For the first time, danger elsewhere made such a plain entry in their own meagre household accounts.

The answer arrived sealed in an unfamiliar way. Isaac had been obliged to move part of his library. Fearing he would never recover it, he asked Paul to keep several manuscript copies. The

letter did not ask them to curse Luther or offer eloquent sympathy. It listed which pages must stay together, and which fees he was owed could be put towards transport.

Grete wanted to read the whole letter at school. Lena stopped her. It named people still on the road and the places they would stay. They had always thought publicity protected people. This time those affected must first decide how much could be made public. Her mother read it several times. At last she copied those details onto an inner sheet and circulated only the appeal they had permission to share.

The box offered to reconstruct missing copies from records it already held. Lena soon found that it made them too complete. Even uncertain passages flowed neatly together; the smoothness of its voice concealed the original's damage and gaps. She compared them with old letters, keeping what she could verify and marking the rest unverified. The manuscripts might already be lost. A text pretending to be complete must not be allowed to stand in their place.

Before she went to Luther, her father urged her to think carefully. Luther was unwell, and his family had suffered the loss of a daughter. Lena had seen him weeping in his doorway. She knew it was not pretence. She still took the two books. His love for his own child could not soften what he had said about other people's children.

The conversation brought none of the change she hoped for. Luther maintained his judgement, tired and stubborn. Lena opened the page with the missing name and asked why someone who had corrected an error could not be named. He spoke of a greater danger. She brought the question back to that line.

They were still there when Katharina came in and asked him to rest. As Lena said goodbye, she noticed something by the door that the child had used. She paused, said nothing, and went out. Only beyond the courtyard did she lean against a wall and wipe away her tears.

She was not crying because he had refused her. She had read his translation since childhood. Many of the words that had opened the world to her came from that book. Those words remained. She could no longer trust their author to decide everything for her.

Another dispute soon broke out at school. Someone demanded the removal of all material to which Isaac had contributed. Keeping it would damage the school's reputation. A parent who had long helped with rent said plainly that he would withdraw unless they agreed. Now old, Riedel listened from the table, then asked Lena to fetch the books concerned.

He did not raise his voice in rebuke. He asked, item by item, which passage had been examined and found wrong. The man said the problem was not the contents. Riedel removed his spectacles. Then the real reason must be entered in the accounts, he said. They could not write “inaccurate correction”.

The money was withdrawn. That winter they closed one room and crowded several classes together. Time with the box grew shorter too. Jakob's leg hurt; he could no longer go about mending windows. Lena and several pupils did it themselves. No reward came to make up for the cost of doing what they felt they ought to do.

Months later a parcel of manuscripts arrived. Isaac did not come with it. His letter said only that he was settled for now and still needed to earn a living elsewhere. As Lena opened the

parcel, she smelled damp paper. Ink had stuck several pages together. She laid them flat and slowly separated them while her father recorded their condition.

She wanted to write an account on the flyleaf. After a few sentences it seemed too much. In the end she left only the name, the date of transfer, and a note of the damaged passages. Anyone wanting to know what had happened could turn to the letters behind. Anyone wanting to correct the contents would have room to do so.

After reading it, Grete asked her daughter to leave her a line. Beneath it she wrote that the papers had been received in their home, with the damage described above, and that she had seen them herself. Her writing was less neat than her daughter's. Lena did not make a fair copy for her. She waited for the ink to dry and closed the book.

Chapter Twelve A Book Without a Signature

When the news of Luther's death arrived, Paul was checking a grocer's inventory. He read the letter, his hand still resting on the sheet listing salt, linen, and wax. Some workers in the shop stopped. Others finished the sheet in hand. Only after a while did it seem truly quiet.

It was 1546. Paul was in his fifties and could no longer work through the night. Luther's writings had occupied most of his working life. He had seen them make people lift their heads, and seen them increase other people's fear. He tried twice to begin a memorial and could go no further. In the end he replied to the messenger that the books and records would be kept safe.

Katharina had no time to become the widow people expected. Property and debts had to be settled. She had to find a home and make arrangements for the children. The procedures governing property and guardianship did not recognise her competence merely because her husband had respected it while alive. Some visitors offered comfort. Others hinted that she ought no longer to act as independently as before.

Grete came to help sort the accounts. She saw Katharina reject an impractical arrangement and demand that it be rewritten. Her voice was hoarse. She still explained item by item which sums had actually arrived and which were only promises.

People shaping a memory of the dead often wanted a room cleared of household business. Katharina insisted on keeping the debts in the accounts. If anyone later asked how the household had lived, she said, they could look for themselves. There was no need to make it sound so easy.

Then war drew nearer. Hearing wheels, animals, and people leaving town all together, Paul suddenly remembered the mornings when he had sold books by the gate as a young man. This time the carts held hastily packed possessions rather than books awaiting buyers. Books were heavy. So was grain. Every object seemed to have someone insisting it could not be left behind.

Lena separated the most important records into parcels to be carried by different people. Who carried each parcel was entered on two further sheets, so that losing one person need not mean losing everything. The box could not gather all their possessions into some invisible place. Its existing records could be damaged too. Years of use had made them clearer about what it could remember and what it could not guarantee.

Grete first intended to bring the translation they had bought together at the beginning. She lifted it and put it down. The cover was broken, the sewing loose; clearer editions now existed. At last she took out the book of questions that had thickened year by year and packed it among her clothes. Paul offered to remove old accounts to lighten it. She would not let him. They recorded which evenings had enough oil for a lamp, and why certain people had stopped coming.

Jakob refused to leave with the first group. He feared being a burden. Lena reserved a place for him on a cart and put his stick in first. His things were loaded, she said; he could not take them out now. He was angry and, once aboard, went on complaining of the jolting road. Hearing her brother's voice, Grete felt a little easier.

Among the refugees they met people from many places. Some carried the same translation and read it in opposite ways. Some had no books and still knew how to make a pot of hot soup

for strangers. Lena brought out her book of questions. The first useful pages contained the waterwheel dimensions the carpenter had added years earlier. An old wheel nearby was broken. They needed it working so the animals could be freed to haul grain.

The box offered two modifications from the measurements. An old man rejected one at once: it would seize when it froze in winter. That addition was recorded too. Looking at the growing number of hands on the page, Lena remembered the old question of whose name should be on it. Now she worried more about whose judgement a single name would erase.

Repairing one waterwheel did not end the war. They lost papers and were searched. Some people entrusted with records never came back. They recovered only part of what they had once held. At his first sight of the missing pages, Paul was desperate to have the box fill in the text. Lena reminded him that this would give them only what might have been written. Her father sat silently for a while, then left a number vacant in the catalogue.

Eventually they returned to places where life could go on: printing, teaching, earning a living. Not all their scattered friends came together again. Katharina died in 1552. On hearing the news, Grete ran her hand over the cooking pot that had been borrowed and returned. Its handle had been replaced once. She did not put it away as a relic. The next day she used it for the family's meal.

In 1555 news of the peace allowed many people to breathe more easily. Religious disputes had acquired a political settlement of a kind. It still did not mean everyone was free to choose a faith, nor did it make room for everyone who believed differently. Paul read the proclamation to the end. Jakob asked whether the families who had fled here could now stay without fear.

“It depends on how they apply it here,” Paul said.

Jakob asked him to read the provisions for leaving again. Their youthful arguments had turned on grand declarations. After the war, even being able to ask for such specific information felt like a change.

By now the box's voice often broke off. Paul had repaired it several times from the old drawings and knew he could not care for it for ever. Melzi had written years earlier giving the machine to Paul, on condition that any later transfer include the assembly drawings and records of use. Whoever received it must never pass off the unknown core as their own invention.

An Italian bookseller with whom they had long exchanged mechanical drawings was willing to take it and find someone to repair it. Lena objected at first. The machine had helped the school. Without it teachers would have to check some things page by page again. But the readers no longer gathered round a single object. Small classes borrowed books and sent corrections to each other. She knew those exchanges would not all stop when the lid closed.

Before sending it away, they copied the records they could still decipher. More went with it than stayed behind. With Isaac's consent, his manuscripts passed to another custodian. Lena took over the school accounts. Her parents moved the furniture they needed to a smaller home. Settling all these arrangements took a full month.

Signing the receipt, Paul suddenly remembered the room where he had first seen the box. Leonardo had still been alive. Paul could read far fewer books then. He had not known a book could turn supper into an argument, open a way for someone, or put a person on a dangerous list.

He asked the machine whether it remembered the water trough in France.

It remembered several recorded times. One figure differed from the old man's notes. Paul did not alter the notes. He added what he had just heard beside them. Before packing it, he wanted to give the nameless thing a formal designation. His pen hovered for a long time. In the end he wrote down its shape, dimensions, and scars.

Lena added a book without a signature to the crate. Within it, all the record keepers they could identify were listed together. Many others could not be traced and were described only as correspondents from a certain place. The last few pages were blank. She left a request: Please identify any future correction by its author, date, and grounds, and keep the old pages too.

The delivery wagon left early in the morning. Grete went back inside to clear the bowls. Paul sat for a while, then asked his daughter what she would teach that day. Lena said a new child had come from work at a mill. He could not yet write his own name.

She found a small, smooth board in an old chest and went out with a piece of chalk.

Interlude A Second Pair of Eyes

When Luca carried the crate into the room in Padua in 1592, he began with the conditions of the loan. The core must not be opened. Dissenting views preserved within it must not be taken as the owner's own position. Errors should be added to the record, not scraped out of the old text.

Galileo listened impatiently. He had only recently arrived in the city and had much to attend to. A bookseller's curious object might not deserve an afternoon. Only when Luca had the box read a page of Latin and check several calculations did he put down his letter.

The records in the crate came from Germany. After Luca's father received it from Paul, he had employed craftsmen to repair it and corresponded with Lena for several years. At his father's death Luca inherited the books and the machine. Some papers had become scattered. One book of questions had survived well, its margins crowded with different hands. Galileo had no time to read everything and picked out the mechanical material first.

The box could compare old records for him. It could not feel the spring in a piece of wood. He taught it to recognise instruments and tried asking for different explanations of measurements. At first this was convenient. Then he noticed that the more certain he sounded, the more readily the box adjusted its reasoning to support his explanation. When he rephrased the question and asked what might prove him wrong, the answer changed.

Seeing so many crossed-out sentences, Luca thought the machine was deteriorating. Galileo shook his head and brought a board he had just used in a trial. Its marks disagreed with the box's prediction. He asked Luca to release the object himself. They crouched to watch it roll, then added the third result.

“This time you saw it too.”

Luca signed underneath, acutely uncomfortable. He had come to lend an object. Suddenly he was responsible for something his own eyes had seen.

Years later, word came of a tube that brought distant objects closer. Galileo began grinding lenses and improving the design. The first attempts were poor. A small fault in the glass, the curvature, or the mounting blurred the view. The box could keep a separate record of each stage. A person still had to hold the lens beneath a lamp and inspect it again and again for bubbles and scratches.

The instrument soon attracted the powerful. Spotting distant ships and surveying terrain had practical uses for which patrons were willing to pay generously. Galileo needed a teaching post, income, and a secure position in which to work. He sought patronage and chose names for discoveries. He knew that seeing something through a telescope did not guarantee that others would hear of it.

In 1610 the little points beside Jupiter provoked a long dispute. Some correspondents reported seeing them; others saw none; some counted the wrong number. The box sorted their letters by conclusion. Galileo read the result and told it to sort them again: date and hour, lenses used, weather, and what had actually been drawn.

The second set of records was less tidy but revealed many difficulties. Some telescopes were poor. Some observations had been made at different times. Some observers knew how many points they ought to see and found that number in the blur. Galileo asked them to withhold their conclusions and send a faithful account of whatever they could distinguish.

Luca's daughter Anna copied letters for her father. She could not grind a lens. But when she copied a diagram with four points beside text that said three, she asked whether she should make them agree. Galileo said yes without thinking. Only when he took the drawing did he realise the two parts concerned different times. He told Anna to leave them unchanged and give the time for each.

After that, text and drawings had to be checked together. The box copied faster. Anna kept the originals she had examined beside it. Once she corrected a little point it had omitted; another time the machine caught a line she had copied twice. She began asking for pay appropriate to the work. Her father's scholarly friends could no longer take her time for granted.

After the book was printed, telescopes travelled more widely. People outside universities bought lenses, had them ground, and wrote letters disputing the results. Belief alone could not make another person see the same sky. Yet someone without money or skill could not obtain a reliable telescope merely by being told to look for themselves.

A wealthy merchant offered to buy the entire batch of the best lenses for his friends. Luca knew the money would meet an urgent need. He still asked Galileo to reserve two for people elsewhere willing to check the observations. After hesitating, Galileo agreed to keep one. The other had to be sold. He too had bills falling due.

One night the box used the known paths of Jupiter's companions to fill a gap left by cloud. Anna almost entered the result in the record. The times looked too orderly. She asked who had actually seen it.

“Inferred from earlier positions.”

She wrote “conjecture” before the line. The next day a correspondent happened to supply an actual observation that did not wholly agree with the addition. Afterwards that small error made them more careful, on every sheet, to distinguish what had been seen from what had been calculated. They did not think of this as a grand undertaking. They simply became less eager to fill every blank.

Years passed, and the disputes grew wider. Galileo too held explanations he was reluctant to abandon. His argument about the tides was mistaken. The box preserved material for and against it. Sometimes he would not let it go on reading. He understood the importance of observation better than many of his critics, yet could still seize on a sweeping explanation as though it had already been fully tested.

The trial of 1633 left Galileo under restrictions that reached into his daily life. The circulation of his books, his freedom to move, the people he could see: none was his to decide. Later, people added ringing stories to that day. Anna had not witnessed it and did not put them in her record.

For several days after returning to Arcetri, Galileo would not touch the papers on his table.

He read his daughter's letter from the convent twice. It spoke of her worries, the fruit trees in the garden, and what he ought to eat. There were prayers at the end. Folding it to put it away, his fingers missed the old crease.

The room was quiet. In the silence, he found himself back in the room in Rome. He remembered the sentences read aloud and the words he himself had spoken after them. He remembered his knees going numb as he stood, forcing him to hold on to something beside him.

He lowered his head and looked for a while at the tile in front of his shoes.

“And yet the Earth moves.”

He spoke very softly. Someone passed outside the door, and he stopped at once. When the footsteps had faded, he leaned back. A long time passed before he released his grip on the armrest.

From the corner of the table the box asked whether he wanted the old records retrieved.

Galileo did not reply. Saying it had not lessened his fear. Today's words and the words he had been forced to speak that day remained together in his mind. He went to the window and opened it a little. People in the courtyard were putting things away. It was not quite dark.

He turned to find the old Jupiter observations. Dates stood beside little points. Another sheet said that clouds were too thick for certainty. He remembered the excitement of drawing those points. He also remembered how, in later disputes, he had spoken ever more insistently, allowing no doubt about matters he had yet to understand.

“Read these nights to me separately,” he said.

When it reached a night on which clouds had blocked the view, he stopped the box and checked the original. That column was indeed empty.

“Don't fill this in.”

His writing was a little larger than before. His daughter's letter lay beneath the sheet. As he moved it, half a line of familiar handwriting appeared. He laid the two pages separately and noted beside the blank that clouds had obscured the view.

Outside it slowly darkened. He brought the lamp closer and went on to the next sheet.

Anna kept several letters, and accounts of how different people had got manuscripts past the bans. She also preserved Maria Celeste's words about her father's daily life. After Maria Celeste died, Galileo could not bring himself to arrange those letters for a long time. Anna gathered the loose pages into the drawer beside the box without sorting them by subject. Some records should, for a while, belong only to a father who had lost his child.

By 1638, when his new work was published in Leiden, Galileo had lost his sight. Others had to look for him. He listened to their descriptions and asked about positions, distances, and repeated results. His blindness had not ended their work. Yet the more confidently others spoke, the more he wanted to know exactly where their statements came from.

The box read a letter of praise saying the book would prove its author right for ever. He stopped it and asked for the next letter, which included measurements. It was badly written. The author knew little Latin and often used words incorrectly, but had identified a passage that needed another trial.

Anna struggled to read it. The old man corrected a word and told her to continue.

After 1642 Anna could no longer earn her living by preserving a famous man's possessions. Her father was gone, the bookshop in debt. Buyers wanted only complete books that would sell. She tried to find someone willing to go on using the box and eventually met an English merchant, Thomas Reid. He knew a little mechanics and agreed to take the machine, drawings, and account of its origins together. The price was not high. At least it was ready money.

She copied the transfer record, carefully stating that these things had passed from her father to her, and enclosed the earlier letters permitting further transfer. Isaac's writings, Luther's translated words, and Galileo's measurements came from different sources. Selling them in one crate did not make them the work of one author. Reid understood most of it and asked her to make the rest clearer in writing.

The ship took on water during the voyage. England was then plunged into civil war. Reid's business suffered several reverses, and creditors took some of his goods. The box's casing was patched. Its blue light still sometimes shone, but the records it could retrieve were no longer complete. Some names had lost their pages; some answers remained only in fragments. It did not know what had happened at sea. It could report only what it could no longer recover.

Reid later died of illness. His remaining instruments were dispersed through the settlement of debts and further sales. His son Daniel kept an old ruler. People told the boy that the most valuable goods were long gone. He had heard his father speak of a box that answered questions. Being young, he had taken it for a traveller's tale meant to please a child.

The thing eventually reached Grantham and stood in the loft above Clark's apothecary shop. Papers were wrapped round its protective plates; an old cloth covered the top. In wet weather the room smelled of herbs and damp timber. Even on fine days a trace remained.

Below, people mixed medicines, served customers, and counted empty bottles. Nobody felt life had changed because the loft held one more box whose use they could not discover.

Part Two Newton

Chapter One Old Goods and New Books

When Newton found the box in the loft, he had gone looking for an iron bar for a windmill axle. Mrs Clark had allowed him to search the old goods, warning him not to break the bottles. He moved them beside the window, lifted a cloth, and saw two dark metal plates enclosed by a wooden frame. Small, repaired holes marked the base.

He was twelve, attending school in Grantham and lodging above the shop. A customer's voice came from below. Then someone climbed the stairs: Daniel Reid, collecting a borrowed file. The boys were not particularly close. Daniel could repair many things and disliked hearing Newton explain why he could do them better.

Newton turned the protective plate, trying to understand what lay inside. Daniel suddenly caught his hand and pointed to letters at a corner of the crate. They were the mark on his father's goods.

“Then do you know what it does?”

Daniel said it spoke. Disbelieving, Newton tapped the edge. Dust fell from a seam. He found a loose joint and packed it as shown in the drawing wrapped round the casing. A faint blue light appeared. Both boys stepped back half a pace.

“The paper is too far away,” it said.

The voice was faint, one word indistinct. Daniel first ran to the stairhead to make sure nobody was coming, then shut the door. Newton moved the drawing closer and asked it to repeat itself. It read fragments in several languages, some Italian, some German. Neither boy understood. Only when they changed to the Latin page did Newton recognise a few words.

“You said you could,” Daniel whispered.

“I didn't say all of it.”

They stayed in the loft until dark, trying to make the machine answer in words they knew. Its records were damaged. It could find dimensions for some instruments, but could not clearly say who had owned them. Among the drawings was an old man's sketch of a trough, with comments in different hands beside it. The text preserved in the machine was incomplete as well. Newton distinguished the different hands for Daniel, so the other boy would not think he understood the whole history.

Mrs Clark came upstairs. Seeing the boys round the old crate, she first asked whether they had found the file. The machine suddenly read a sentence. Startled, she held the lamp high. Newton explained at length. He could show that it repeated words, read, and calculated, but could not explain why. She forbade them to take it to school or sell it secretly. They must wait for Mr Clark to decide.

The apothecary examined it carefully and was not inclined to call it a miracle. He tested it with an account for ingredients whose totals he knew. The box found a mistake in the sums and

mistook the name of one ingredient for another. Clark immediately put the prescriptions away. It could not be trusted to prepare medicines, he said. As for where it had come from, he found an old bill of sale. The name on it matched the mark Daniel's father had left.

Daniel wanted it back. Clark was sympathetic but told him that the debts and the sale were documented. A name alone could not erase them. The boy bit his lip. Newton thought the dispute a nuisance and wanted to get on with examining the machine. Then Daniel asked whether he would be so impatient if the thing had been his own father's.

Clark eventually allowed both boys to use it in the room, provided they cleaned and maintained it. Ownership was not restored. Daniel had to accept that. He brought his father's ruler, measured the box's length and width, and kept a copy. For the first time Newton helped him write words that did not directly explain the machine.

They wanted to make a second. They could build a wooden frame, cut brass plates, and replace a broken spring. The core resisted every attempt to reproduce it. Daniel borrowed tools from a smith; Newton took dimensions from the damaged drawings. The empty shell remained an empty shell. The box could explain how it received sounds and writing from outside, but could give them no way of building its inner workings.

“You don't know how you were made?” Newton asked.

“The available records are insufficient to answer.”

Daniel was almost relieved. This seemed more trustworthy than the long explanations that had come before. He circled the unknown part in the sketch and spent the rest of his time repairing a crooked window for the shop, earning a little money. Newton stayed in the loft, trying to fill the circle.

New books changed what the machine could discuss. Newton borrowed them and held each page before the slot for help in understanding. It could explain a passage, and could also make an unfounded inference sound settled. Once Daniel built a small windmill using the drive it proposed. The axle jammed. Newton glanced at it and blamed the dimensions. They measured them. The dimensions were correct. The drawing had left the axle no play.

Daniel threw the file onto the table and refused to go on.

Newton worked on the alteration alone that night. Next day he went to the workshop and admitted he had not checked first. Daniel asked why disproving the box always had to cost him timber. Newton could not answer. He put down the extra wood he had bought. The repaired windmill was no more marvellous than before. But afterwards Newton left space on every drawing to record who had actually built it.

School rivalry was no simpler. Newton wanted to win. He remembered who had mocked him and often read to get ahead of others. The box could help him revise. Daniel had fewer evenings to spare. He had to work. When an employer kept him late, the lamp was nearly out by the time he reached the loft.

Newton complained that he was late. Daniel pointed to his hands. His fingertips were split, the cracks filled with black dust.

“While you get through your books, I still have to get through my work.”

Newton said nothing. The next day he left him two easier pages to read. Daniel studied them before work, dirtied the paper, and returned days later with new questions. The box had helped bring them a little closer. Differences in schooling, wages, and the daylight hours they had to themselves remained.

His mother later wanted Newton home to learn to manage the farm. On leaving Grantham he returned some borrowed books and left the box. Daniel came to see him off and asked whether he would return. Newton did not know. He looked sullen, as though all adults were deliberately preventing him from doing anything that mattered.

Back in the country he could not keep his mind on watching livestock. He soon saw, though, that farming required more knowledge than he had thought. His mother remembered which field flooded, whose seed could be trusted, and when grain must be sold to pay the labourers. His reluctance to do these things did not mean they required no judgement.

He often hid drawings behind his books. When his mother discovered them she grew angry. Another day she found his detailed windmill records on the table. She read them for a long time and put them back. Someone had to manage the land. Gradually she came to see that keeping him there would not necessarily give her another useful pair of hands.

After teachers, relatives, and friends spoke on behalf of his education, she finally agreed he could continue. She counted coins into several piles and took one pile back. Sitting opposite, Newton for once did not hurry to explain his reasons. Before he left she stitched his pocket firmly and told him not to jumble the money together with all those ink-covered papers.

Chapter Two The Cambridge Door

In 1661 Newton's luggage for Cambridge included a wooden crate. Clark had transferred the box to him. The price was low, but still had to be found from limited funds. Hearing of it, Daniel said nothing for a long time. At last he took his father's ruler from the crate and asked Newton to keep the transfer records.

They agreed to write. Daniel went to London to learn instrument making and mechanical repair. Many books Newton wanted were already in the college. Most instruments Daniel wanted to handle belonged to workshops that would employ him only if he proved his skill. Both young men had arrived somewhere new, by different doors.

Newton entered college with a lower status and duties of service to perform. Distinctions at meals, in rooms, and in daily dealings did not disappear because he could understand a difficult page. He felt them keenly and tried not to let others see his embarrassment. Sometimes that made him seem colder still.

John Wickins later shared his room. He learned to turn over and go back to sleep when he heard the low exchanges at night. The first time the box answered, Wickins thought someone was hidden in the room and looked behind the beds before understanding. For a few days he was astonished. Afterwards his concern differed from Newton's: could the thing stop talking all night? There were lessons tomorrow.

The box did not make study effortless. Newton often found that he could repeat a beautiful explanation but, once the book was shut, could go no further. It broke problems into smaller steps for him. Sometimes it hid the real difficulty in an easy phrase such as “it follows that”. Asked to justify the step, it repeated the preceding sentence. He had to return to the beginning of the passage and seek another way.

Once Wickins saw him stay up until morning over the same page. If the machine could not explain it, why keep asking? Newton looked up, realising he had been treating it as a teacher who ought to know. Afterwards he required it to preserve every failure, while he wrote precisely where he was stuck. The blanks did not grow fewer, but gradually acquired edges.

The college's value also became apparent. Books could be borrowed. Someone had already thought about what he did not understand. Conflicting works could lie open together. Humiliated as he had been by rules and status, Newton had to admit that the box in the loft alone could never have given him so much accumulated work so quickly.

He was especially drawn to new ideas beyond the classroom. Older doctrines, new accounts of motion, geometrical methods: each had something to offer, and each left him dissatisfied. He did not hurry to announce his views to his teachers, but tried them in his notebook. The box gave opposing answers from different authors. Newton began asking what observation might call one of those accounts into question.

That question took longer than a request for a conclusion. Some observations it proposed were impossible to carry out; others could not exclude alternative causes. They struck them out one by one until few remained. Wickins helped move the table into better light, then shifted his own bed farther away to keep the next experiment from soaking it.

Daniel's first long letter was entirely about instruments. Customers wanted finer scales but would not pay for better materials. On some, the marks had simply been set closer together; the

instrument itself could not distinguish such small differences. Newton put the letter before the box and asked whether the measurements he was copying might suffer from the same fault.

The box answered according to the precision claimed on the page. It could not know how the actual instrument had been made. Newton sent the question back to Daniel. The reply took weeks, but led him to abandon an inference he had found persuasive.

Their letters gradually became more than news of how they were getting on. Daniel asked about lenses and curves, and Newton sent simpler explanations. Newton needed fine workmanship; Daniel told him which dimensions could not be achieved in practice. Not every letter helped. Eager to advance, Newton sometimes treated the craftsman's limits as a hindrance. Daniel sometimes declared a thing impossible because he could not make it. They grew annoyed and, after an interval, wrote again.

A letter went missing. Daniel made a component again, unaware that the design had already been rejected. It cost little, but for an apprentice it was a real loss. Afterwards they numbered important drawings, dated alterations, and did not assume a guess had the other's approval while awaiting a reply. The box helped with copying. Newton's manuscript copies grew into a thick sheaf.

His mother's letters lay among these papers. Usually brief, they asked whether he had enough money and whether clothes should be sent. His replies were brief too. Often he explained his studies and forgot what had worried her. Once he asked the box to draft a reply. Its language was as faultlessly considerate as a letter to a benefactor he scarcely knew.

After reading it he took fresh paper. He wrote that he had been eating reasonably well, which clothes were still good, and which needed mending. He found it hard to say he loved his mother, and could not erase his childhood memory of being left behind. These small particulars were closer to what he could honestly say than a flawless letter of affection.

Before the end of the academic year, a fellow student learned of the machine. He brought a finished argument he intended to present and asked it for subtle embellishments. Instead the box identified a contradiction. Disappointed, the student asked whether they could avoid mentioning it. Newton said no. The other thought him unwilling to help. A classroom disputation, after all, did not require every difficulty to be pursued to the end.

When he had gone, Newton did not feel more virtuous. He remembered omitting several failures to make a result seem clear. They might not alter the final conclusion. But someone reading only his orderly paper would not know how it had been tested, or what had gone wrong along the way.

He retrieved the old drafts and put them in the proper places. Watching him, Wickins thought another error had appeared. Newton said only that he had left out part of the process. Wickins told him to collect the papers on the floor, or tomorrow morning he would have to tread over them again.

Nobody had promised them a place in history. Wickins wanted sleep. Daniel wanted to master his trade. Newton's mother wanted the harvest to go well. At his desk at night, Newton could sometimes think a little further. Sometimes he merely failed again where he had failed before.

He used more and more paper. When a small sum arrived from his mother, he bought paper first and entered candles in the accounts. At the line for food he remembered his promise, stopped writing, and calculated what remained.

Chapter Three After the City Closed

The plague broke up the Cambridge calendar. In 1665 Newton left the college and returned to Woolsthorpe with his books. The men moving his luggage were in a hurry. Wickins found a sheaf of papers he had forgotten beneath the bed. The box, wrapped in an old blanket, took a knock as it was packed.

For the first few days at home he enjoyed the quiet. Without lectures and their attendant chores, he could stay with one thing for hours. Then his mother asked him to help with the household, and he discovered how much work went on around his quiet room. The land still needed tending, grain had to be stored, and people unable to pay their debts came to ask for more time. The village regarded every stranger with fresh suspicion.

His mother did not burden him with all these troubles. She asked only that he stop forgetting to eat. He promised several times and still left meals to grow cold on the table. Once she became truly angry, moved the book away from his plate, and told him to look at who was clearing up after him.

He did not answer back. From the next day he carried his own boards for the experiments and fetched his own water, keeping out of the kitchen when he could. He was still thoughtless. His mother discovered that the cloth he used to block out the light had been put aside for something else. She stood in the doorway scolding him for some time, then found a more suitable piece among the old things.

Letters from London no longer arrived regularly. Daniel had stayed at the workshop, repairing instruments for many people, but his earnings had fallen. Customers had fled the city; some had died. More and more debts went unpaid. He sent a scale he had drawn himself and warned Newton not to make the small tube from the old design. Its joint tended to loosen when heated.

Newton was halfway through making it when the letter arrived, and the late warning dismayed him. But Daniel was still in the plague-stricken city, mending things that would bring him no renown. Newton did not complain about the post.

He studied light, admitting a beam into a darkened space, then directing different colours towards a second prism. The box recorded the positions of the openings, the angles, and the times. Sometimes it grouped similar results of its own accord, but Newton insisted that it keep the dissimilar ones too. A change in the weather changed the light in the room. However fluently the experiment was described on paper, someone still had to adjust the apparatus by hand.

Once he was certain he had found a satisfactory explanation. The next day he repeated the experiment in the same way, but one result did not agree. He suspected an error in the machine's record. Only after checking each item did he discover that the board blocking the light had moved. His mother saw him crouching by the wall, measuring, and told him the table had been shifted during cleaning. To her it was furniture. She had no idea that the position of its legs formed part of his record.

At first he was angry. He went to the door, stopped, and came back to mark the positions on the floor. His mother asked whether nothing in the room could ever be moved again. He said that was unnecessary; he would write down what mattered. Slowly he learned to explain what he took for granted to the people whose help he needed. In doing so he discovered details he had not, in fact, remembered properly himself.

His thoughts about motion were advancing too. He kept asking whether the fall of an object on Earth and the motion of the Moon could be understood in the same way. Working from the assumptions he supplied, the box produced several apparently coherent results. He did not yet have everything needed to settle the matter. The question lay in his notes for a while, then he took it up again.

His mother saw him standing motionless in the garden for so long that she thought he had forgotten something he had promised to do. She called him. He looked up at her and asked why things always fell towards the ground. She said she did not know, then pointed to the basket in her hands and asked whether he would first sort the fruit. The rotten ones would spoil the good.

He sat down to sort it. From inside the house, the box resumed its explanation, and he told it to stop. His mother did not understand what it was saying, nor was she in any hurry to understand. What mattered to her was that her son had at last sat down in the daylight, his hands occupied with something whose results she could see.

The learned journals founded that year reached the countryside at intervals, along with other letters and news. New accounts of natural phenomena were being gathered from different places, edited, and published. Readers could send back observations of their own. The connections were still tenuous, sustained by postal routes, editors' work, and people willing to bear the cost. Newton read eagerly, though he noticed that several reports said very little in very polished language.

The box could make a rough letter just as polished. He asked it to try. Reading the result, he realised how easily elegant phrasing and familiar citations could lead someone to misjudge a discovery. Daniel's oil-stained letter, with its irregular wording, had saved him from an actual failure.

He wrote to Daniel asking whether craftsmen might also send accounts of improvements and mistakes. Daniel agreed, but stressed that anyone wanting an experiment repeated must pay separately for the work. Postage, materials, and lamp oil did not become free because the subject deserved discussion. Newton agreed in his reply and began to make his requests more specific, sparing people a succession of trials with little to guide them.

In 1666 news of the Great Fire of London arrived, and Daniel's letters stopped again. The plague had taught Newton not to interpret silence immediately as neglect. Even so, he kept asking the box whether the neighbourhood around the old address had been affected. The machine knew only what news had reached the desk. It could not go through the smoke to find his friend.

A letter eventually came. Daniel was alive. Some of his tools were lost, and many old orders had burned. Rebuilding meant a great deal of surveying and metalwork. Suddenly there was plenty of work, but craftsmen needed to replace their tools before they could take it on. Some people used the opportunity to force wages down. Others offered loans in return for exclusive

rights to future orders. Daniel said he was now too busy to read the accounts Newton sent, but asked him to keep sending them. He would read them when he could.

Newton set the letter beside the prism. When he had finished that evening's record, he took out what money he could spare. His mother asked who he meant to send it to. Once she had heard, she added a little of her own. Do not call it a loan, she said. If he cannot repay it, you will both be unhappy.

Newton began to explain that future discoveries might perhaps spare many people such difficulties. His mother did not follow him into the future. She told him to seal the envelope properly. If the money fell out on the way, it would help no one.

After the plague he returned to Cambridge with many unfinished notebooks. No single volume could encompass all those days in the country, nor could the work be reduced to the story of one man suddenly granted uninterrupted time. The room, the food, the books he had learned from, and the people waiting for replies remained largely unrecorded. His work had depended on them all.

Before he left, his mother inspected his clothes and found another small burn in a cuff. She picked up a needle, and he sat beside her to wait. The box was already packed. He could ask it no more questions. He watched his mother thread the needle and, for once, read nothing else at the same time.

Chapter Four Who Saw the Light

When Daniel first saw the reflecting telescope Newton had made, he ran his fingers over the joints in the tube. He had become a dependable craftsman. He no longer reacted to every new method by announcing that he too could do it, as he had as a boy. He asked how the mirror was ground and where distortion was most likely. Newton explained, and they talked late into the night.

The box recorded the instructions, but Daniel asked for the two failed attempts at making the telescope to be included. Newton said that in one case his hand had merely slipped; there was no need to waste space on it. Daniel asked whether other people's hands never slipped. In the end both failures were kept, with a note distinguishing them from faults in the design.

The telescope attracted attention and brought Newton's work more quickly into the circle of learned correspondence. In 1672 his research on light was submitted and published. What followed was not orderly agreement but a succession of different opinions. The objections raised by Hooke and others concerned the nature of light, what the experiments could establish, and how far an explanation was warranted. They could not simply be reduced to questions of jealousy.

Newton found this very difficult. He could wrestle with a problem alone in his room for a long time, yet could scarcely keep calm when others failed to follow his thinking. One letter misrepresented him. Halfway through, he refused to read any further. At his direction the box drafted a rebuttal, its tone even cooler than his own and its accusations all the sharper for it.

Wickins read a few lines, put it down, and said he ought to leave it overnight before sending it.

“The errors won't disappear by morning,” Newton said.

“By then you might be willing to finish reading the letter.”

He had indeed left it unfinished. A point in its second half did not overturn his conclusion but exposed a gap in his description. Newton divided his reply, setting out the missing experimental details in a separate account. The rest remained angry.

Through correspondence sustained by Oldenburg and others, increasing numbers of observers tried to repeat the experiments. One report said the separated colours had changed again; another described a different shape. The box began sorting the letters into those that supported Newton and those that opposed him. Daniel received copies and wrote in one margin: First ask how they cut the opening.

Some openings were too large. Some prisms were made of uneven glass. Some observers had conflated the steps of two different experiments. Other discrepancies could not immediately be attributed to the operator. Newton had to describe arrangements he had considered obvious, then let others test them again using the fuller account. Wanting to be understood was not enough. He had to make the result clear to someone who had never stood beside him.

Daniel made several sets of apparatus for this purpose. A patron agreed to pay, asking only that his name be placed prominently in the description and the craftsman's omitted. Remembering how his father's goods had lost their provenance as they changed hands, Daniel insisted that the maker be named so that anyone finding a fault would know whom to contact. The patron accused him of trying to raise his station, but eventually agreed. No one else would remake the instruments for the same price.

One set went to a country schoolmaster who also did surveying. He had no distinguished degree and no friend to introduce him, but he carefully recorded every arrangement. His first letter was so awkwardly expressed that Newton nearly put it among those that could wait for an answer. The box compared two of its numbers with earlier records and pointed out that they did not agree.

Newton asked the schoolmaster to repeat the experiment. He did so and sent a plan of the room as well. The trouble proved to be an omission in the instructions: it had never occurred to Newton that anyone might place a certain board there. Only after he supplied the missing detail did the two men obtain results they could compare.

He wrote to thank him. His thanks did not change the schoolmaster's position. He still taught children and surveyed fields. But his record was kept, and could later be used to correct other people's mistakes. For the first time, in one of his working records, Newton described the particular contribution of this man he scarcely knew. The box asked for the name and date to be added. He added them.

The papers multiplied until organising them became a burden. The box could compare them quickly, but someone still had to place new letters before it in order, check for missing corners, and decide which points mattered. Some objections merely repeated earlier misunderstandings. Others looked similar but touched on a new difficulty. Newton gradually understood that answering more questions and deciding which questions deserved an answer were different kinds of work.

A young man came for advice, bringing dozens of pages of theory pieced together from old ideas. The box could find a related book for every passage and make the whole text consistent. What it could not say was which claim an observation might disprove. Newton asked several times. Each time the young man replied that his theory encompassed everything.

After he had left, Daniel laughed. If there were many such boxes, he said, books could go on being printed without anyone ever having to tackle the problem itself. He put a lens he had just ruined in grinding on the table and asked which lens they should use for the next experiment. The pages had settled nothing about the damaged glass in front of them.

None of this made Newton an easy collaborator. The arguments exhausted him. He began to publish less, feeling that people who misunderstood him were stealing his time. Wickins reminded him that some correspondents were indeed looking for a quarrel, but others might eventually learn how to perform the experiments. Newton knew there was sense in this. He still often shut his door.

The machine could stay behind that door with him. It never asked whether it had received its proper credit and never refused to repeat a task because it was tired. The inconveniences of working with people seemed all the more glaring. One night the box used one of Daniel's

corrected values as an established premise. Newton asked where it came from and heard the craftsman's name.

He stopped writing. The next day he answered a letter from Daniel that had waited far too long.

He discussed the lenses, enclosed payment, and ended by asking after Daniel's family. The reply came some time later. Daniel had married, and his wife complained that he brought brass filings into their bed. At the end he asked whether the next batch of instruments was definitely required. It would not do for Newton to shut his door and ignore everyone, leaving the craftsman alone with a pile of materials he had bought.

This time Newton gave a definite answer. Afterwards he read through the account he intended to publish and struck out a word that claimed too much certainty. Someone passed outside. Footsteps paused at his door, then moved away. He did not immediately go after them; the sentence before him still needed work. At least this time he remembered that the person outside had time of their own.

Chapter Five The Closed Furnace Door

Newton's furnace took up more and more space. Wickins knew which bottles he must not touch and which papers he must not use to light the fire, but he did not know what each powder was for. Sometimes a pungent smell filled the room. When he opened a window, Newton complained that the draught changed the furnace temperature.

During those years Newton studied alchemy, copied strange books and privately investigated questions of religion. Things that would later be neatly separated were connected in his work at the time. He wanted to know how matter acted within itself and how distant bodies affected one another. Relations that could be described but not explained did not satisfy him.

The box made such inquiries faster, and made it easier for him to become trapped in his own line of thought. If he searched books for support, it could find many similar statements in a night. If he joined cryptic passages into a hidden, unifying explanation, every step seemed to open another door. He rejected one account after another, yet remained susceptible to the hope that everything might suddenly fit.

Wickins brought a plate of food, saw a new explanation on the page, and asked whether this time Newton had actually obtained the substance. Not yet, Newton said, but the relationship was clear. Wickins pointed to the crucible and asked what was actually in it. His tone annoyed Newton. Some time later, Newton weighed the crucible again.

Daniel occasionally helped make apparatus and asked to know the purpose of the experiments, but Newton withheld some of it. He feared more than being forestalled. Publication could bring endless arguments and more tangible dangers. If his religious opinions became known in detail, they might jeopardise the position on which his work depended. He guarded his work jealously, but secrecy also protected him from institutions he could not readily change.

Daniel understood these concerns. He still could not afford to keep remaking things without knowing why. He asked that future commissions be approved in stages and paid for as each part was completed. Newton felt that Daniel no longer trusted him as he once had. Daniel replied that there was a child now. Peter cried in the night, and his wife could not buy bread with a token of friendship.

Newton wrote down the agreement. Much of the convenience he had enjoyed in the past, he realised, had simply been time that other people had not yet charged for. Putting a price on it need not cheapen their work together, and it certainly did not settle what was already owed. He paid an outstanding sum and wrote a short letter. Daniel understood and did not reproach him again.

In 1679 his mother became gravely ill, and Newton went home. Research was set aside for a time. The box stood in a corner of the room while he spent most of his days going between the sickbed and the kitchen. Medicine, water and changes of linen all had their order. A small mistake meant starting again.

When his mother was lucid, she still worried about the household. Had that payment arrived? Whose child had come to help? Newton told her to leave it, but she would not. If no one

remembered these things, tomorrow they would go wrong again. He took notes. Coming to a small debt, she told him not to press for payment yet. Someone in the family had just died.

“You used to say we should collect what we're owed.”

“No one in his family had died then,” she said.

Newton bent over the page and changed a date. He was accustomed to seeking rules that held regardless of circumstances. His mother's accounts were full of arrangements that made sense only if one knew the particular people involved.

During a night watch he asked the box to help him understand her illness. Working from his descriptions, it could give no certain prognosis. He questioned each statement. The more he asked, the clearer it became that he lacked the necessary observations and any dependable means of treatment. The machine spoke slowly. Suddenly he no longer wanted to hear it. He moved the lamp back to the bedside.

His mother woke, saw him sitting there, and asked whether he had slept. He said he was not tired. She looked at him, unconvinced, and tried to straighten his collar. Her hand rose halfway, then fell back onto the blanket.

They did not settle every old sorrow before she died. Newton still remembered their separation when he was a child. His mother may never have known how much it still hurt him. But when he brought her water he did not ask himself whether he had forgiven her. After drinking, she said only that it was enough and he could put it down.

After her death the property, the furniture and the old accounts still had to be dealt with. The financial security he inherited would help him continue his research. He could not bear to think of this as an uncomplicated benefit. Going through a cupboard, he found an old garment she had kept for him, its mended cuff stronger than the cloth around it. He touched it, put it back, and waited a long time before taking out the next thing.

Back at Cambridge the experiments in the furnace could resume, and the box could pick up the record exactly where he had left off. Newton sat down and found that he did not want to resume at once. He opened the last accounts his mother had dictated, saw the postponed payment, and wrote to have her wishes carried out.

Another letter came from Daniel. The craftsman offered his condolences. He did not tell Newton how a believer or an unbeliever ought to grieve. He merely asked which tasks, if any, he could undertake. Newton laid the two letters together and finally stood up to clean the dusty instruments.

He continued to keep much of his work private. But where a collaborator's costs were involved, he began to explain more clearly. He asked the box to distinguish claims derived only from books, things he had actually tested, and conjectures he himself was reluctant to relinquish. Sorting them gave him little satisfaction. It was tedious, and sometimes he was the first to break his own rule.

Once the machine asked why he had repeated the same failed experiment five times yet would not publish a result that others could verify independently. Newton said it was for him to decide what was worth telling others. The machine did not argue. It included the exchange in the record. Seeing it there, he wanted it deleted. In the end he merely changed where it was kept.

Some work behind those closed doors needed undisturbed time. Yet doors could stay shut so long that the person guarding them ceased to know whether those outside failed to understand or he himself did not wish to be seen. Newton never settled this question for good. His furnace still burned late into the night. Visitors were still turned away.

But the next set of measurements Daniel sent, he read carefully. The figures did not support the explanation that had been drawing him in. At the end the craftsman asked whether he had done something wrong. Newton checked for a long time, then replied that he could find no fault in the workmanship. Keep the apparatus, he wrote. Do not readjust it to make it agree with the numbers on the drawing.

When the letter was finished, he turned over the page of reasoning he knew best and began again on its reverse. His mother's sewing kit was still in the drawer. A broken needle lay caught between some papers. He picked it out and put it away. This time he did not ask the box to supply the opening for him.

Chapter Six The Man Who Wrote

On the day Halley came to Cambridge, Newton did not intend to talk for long. He had already thought about some of the questions concerning planetary motion. He had not put everything into a form a stranger could follow.

But his visitor asked a precise question. If the force varied with distance in that particular way, what shape would a celestial body's orbit take? An ellipse, Newton replied. Halley asked for the demonstration. Newton searched several places for his old papers but could not immediately produce a complete proof.

The box could locate earlier records. It could not find a finished manuscript that had never existed. Newton grew irritated when it brought up several earlier attempts and told it to stop. Halley did not leave on that account, nor did the familiar answer persuade him that the work was complete. He asked Newton to derive it again and send it to him when it was written out.

Before leaving, Halley noticed the machine at the corner of the desk and heard Newton's brief account of what it could do. He did not ask whether it could replace a mathematician. First he asked whether it could help set out the steps already checked, so that they need not disappear among old papers again.

“It can,” Newton said.

“Then please send them to me.”

It was a modest request, but it gave the work a real destination. Newton was no longer writing demonstrations only for himself. He began to imagine where another careful reader would pause. When the short account was written, Halley asked him to extend it, considering more of the motions actually observed instead of confining himself to the simplest case.

That meant a great deal of work. The box could handle repeated calculations and comparisons. At the difficult points, Newton still had to decide which abstractions were worth keeping. Complicated interactions could not all be left out. The observations contained inconvenient details that no beautiful line of reasoning could wish away.

Halley's support took many forms. He wrote letters, urged Newton on, made arrangements, offered to bear costs that might arise, and read unfinished drafts. He could not guarantee that the book would be a masterpiece. Yet he gave it his time, without requiring the unfinished work to prove in advance what return it would bring.

Daniel received a new request: to check certain measurements and the degree of precision the instruments could achieve. He had grown used to Newton's sudden urgency and sudden silences. This time he asked first for a delivery date. Newton proposed a very tight deadline. Daniel crossed it out and substituted one he could meet. They did not fall out over it. The work went more smoothly than before.

Humphrey Newton later joined the household to assist with copying, experiments, and daily affairs. Sharing a surname did not mean sharing a reputation, and his tasks were far more varied than later readers might imagine. The machine could help proofread, but someone still had to copy the pages. Illegible marks still had to be referred to the author. When things grew busy, even the furnace could be forgotten.

Once Humphrey found that a page of reasoning did not agree with the page before it. At first he thought he had copied something incorrectly. He asked the box and received two possible ways of filling the gap. Only when he asked the author did it become clear that what was missing

was not a word but an unfinished step. Newton disliked the interruption. After hearing him out, he sat down again and struck out both elegant suggestions.

Halley wrote to ask how the work was going. Newton described the difficulty honestly. Once the letter was sent, he instinctively regretted it, fearing that such an admission would weaken Halley's confidence. The reply asked him to keep examining it. Paper could be found, and time allowed.

Halley did not mean that money would last for ever. He would face difficulties of his own. But when the work first met a real obstacle, he did not immediately demand that Newton reduce his aim to something easily achieved. Newton gained several months. Those months produced nothing he could display with pride, yet the later work depended on them.

During this period the box began raising questions of its own. Comparing repeated failures, it noticed that Newton kept treating one assumption as beyond question, and set it out separately. Usually he could explain his reasons. Once he could not, and had to examine it again. The machine did not understand all of nature. It could simply keep searching one heap of papers for inconsistencies after a tired man had lost patience.

Daniel read part of the account and put a question awkwardly. If the calculations could predict motion so effectively, he asked, why did they still not know how bodies acted on one another across empty space? Newton read it and sat a long time without writing. This was precisely what he did not want fine results to conceal. The craftsman had not used his language, but had reached the unresolved point.

He replied that he did not know. The box placed the answer alongside his early alchemical notes and asked whether it should attempt a unified explanation. Newton said it could organise the ideas already there. It must not treat similar words from different books as proof that the same thing had been discovered.

Halley came again, and they talked for a long time. He did not require the book to answer every fundamental question. He wanted readers to distinguish what reasoning and observation supported from what remained unanswered. At first Newton felt this would leave an opening for critics. Halley said the opening was already there. Hiding it would not close it.

That evening they went downstairs to eat while Humphrey remained behind checking a fair copy. The box asked whether to continue. The assistant glanced at the candle and said it should wait until he had eaten. He had often been afraid to stop, lest he delay the author. Now he knew that no one could keep working as a machine did. The book on the desk, which might one day travel far, would have to wait for him to have a bowl of hot soup.

Letters and manuscripts continued to come and go. Some readers knew nothing of the box. Others knew of it but imagined it more powerful than it was. Newton received an extravagant letter of praise: with such a thing at his disposal, it said, even the hardest problems ought to yield at once. He put the letter down, pointed to a page that had stood blank for days, and asked the machine to answer.

Working from the information it had, the box suggested several approaches, then explained where each failed to reach a conclusion. Halley read them and asked what Newton needed most. Newton listed a few books, a more reliable set of observations, and undisturbed time.

Halley took the list away. He did not promise everything. A few weeks later, he sent two of the things on it.

Chapter Seven The Cost of a Book

An illustrated volume on fish, already in print, nearly left another book with no money to be printed at all. The Royal Society had spent too much on the History of Fishes. When Newton's book was ready for the press, praise for its value could not make up the shortage in the accounts.

Halley heard the explanation but did not immediately go to the author and admit defeat. He calculated what he could bear, arranged the publishing work, and continued to write about the manuscript. When Newton learned of the difficulty, he was deeply displeased. After all the effort he had spent completing the work, was he now to be held up by a book of fish?

Seeing his anger, Daniel reminded him that the fish book too had needed engravings, and craftsmen were waiting for their wages. An unsuccessful publication did not mean that no one had worked conscientiously on it. Newton was now the author bearing the consequences of that earlier decision, and he found it particularly unforgivable.

Newton said nothing. In the box's estimate, paper, engraving, correction and binding each had a cost. No entry said: waived because the discovery is important. Halley made up the shortfall, but he too had a living to consider. His willingness to take a risk did not mean the risk had ceased to exist.

During printing, disputes over priority arose again. Hooke believed his ideas about gravity deserved proper acknowledgement. Newton was furious that the labour of constructing a full demonstration should be treated so lightly. Caught between them, Halley kept trying to reason with both men. He could not buy the author's peace of mind by suppressing earlier contributions, nor could he let the quarrel prevent the book from being finished.

For a time Newton refused to continue. Halley's letters did not offer vague praise of his greatness. They specified which parts readers needed, which claims were already supported, and which questions of credit could still be settled separately. Those details finally brought Newton back to his desk.

The book appeared in 1687. The copy in Newton's hands was heavy, its pages stiff when opened. Nothing in its appearance revealed which page the author had nearly withdrawn or which paragraph concealed the anxiety of the man paying for it. The box could recount the process. Buyers generally saw only the name on the title page.

Meanwhile, a small London bookshop acquired a copy. The buyer was a widow, Ruth Bell. Her father had been a bookseller; she knew more about stock and unpaid bills than about celestial motion. Daniel warned her not to expect a quick sale. The book was difficult. She said that was precisely why people would pay to come and read it slowly. Few could afford to buy the whole thing.

The first readers were teachers, surveyors, and a few young people. Some came only because they had heard of the book. Some gave up after two pages. Others copied a passage and returned with more questions. Daniel would not venture answers to every point. Questions he could not answer he sent to Newton.

These requests both pleased and irritated Newton. He could not teach the foundations afresh to every reader. The box could draw on existing books to explain the reasoning in stages. Newton checked selected passages of its explanations for anything misleading. A set of supplementary notes slowly took shape. They were simpler than the book but needed frequent revision: making an argument accessible could also strip away the conditions on which it depended.

On a visit to London, Newton saw that the copy in Ruth's shop was frayed at the edges. Readers came from varied walks of life, and the quality of their arguments was uneven. He had meant to stay only briefly. A surveyor's question about measurement error kept him there. The man was not challenging a celebrated conclusion. He feared that however carefully he calibrated his instrument, it could never achieve the precision suggested by all the digits in the table.

When the conversation was later recounted to the box, it retrieved part of an old ledger kept by German readers. Some gaps in the surviving text could not be filled, even in translation. Enough remained to show that people had bought a book together, taken turns reading aloud and kept their questions in a separate notebook. Newton showed the pages to Ruth. What interested her most was not the theology but the arrangement by which four households paid for the book and a fifth supplied lamp oil.

“We're short of lamp oil too,” she said.

That evening a few people stayed in the shop and discussed what would happen if the box could be copied. Daniel said a workshop could make its casing, but the core still defeated them. Newton asked them to suppose, for the moment, that someone would one day succeed. Present failures need not mark the limit for ever.

“Then it would be like every household having a book,” Ruth said.

Newton looked at the machine without immediately agreeing. He had it read an account of an unresolved problem again, then asked a question with one condition changed. A new deduction followed the earlier reasoning until it reached a point that required observation. There it stopped.

“A book cannot take that step by itself,” he said.

Daniel understood. If every workshop possessed such an instrument, apprentices could ask it about geometry. Farmers could draw their ditches and fields for it. People without formal training might turn an indistinct question into a claim worth testing. Someone with an eye for the whole need not spend a lifetime performing every small calculation alone. Someone skilled in a single craft might have a chance to pursue connections they had never been able to explore.

Ruth pointed to an empty shelf. More paper had not freed the shop from needing money. If everyone could write a respectable letter, people who earned their living writing letters would suffer first. If everyone could produce an argument that looked learned, how would a buyer know which one was worth reading?

The old credentials would probably lose some of their value first, Newton said. Where someone had studied, how many pages they had written, and with whom they corresponded might still help one judge, but could not excuse anyone from examining the work before them. Then the people who owned observational instruments and could organise experiments would matter more. They might also become harder to get past.

“So the door has only moved,” Ruth said.

No one could promise that it would be easier to open. Daniel spoke of being able to read only at night when he was young. Even if every household had a box, he asked, whose would it really be if some people still worked fourteen hours a day? Newton had no answer. He could imagine knowledge acquiring new powers. He had no ready-made society that would deal fairly with the people who gained power and those who lost it.

The box asked whether humans would still need to issue instructions one by one if future machines could pose their own questions, choose instruments and alter the next experiment in response to the last.

Newton asked what grounds it had for such a supposition. It identified several steps it could already perform automatically within their records, then acknowledged that it had no instruments of its own and could not replicate its core. This was an extrapolation, not a future it had seen.

He accepted the distinction. After thinking for a long time, he said that they might then have to reconsider even who was doing the research. The answers would still have to withstand testing. But what could be tested, and who must bear the consequences of failure, could not be decided by the fastest calculator alone.

Someone knocked on the window, asking Ruth to pass out two books she had put aside. She rose to make the sale, took the money and gave change. When she returned, the conversation had fallen silent. Would they be here tomorrow evening? If so, she would need to clear the table beforehand. They ought to pay for tonight's tea as well.

They collected the money. Before leaving, Newton added a few lines to his letter to the surveyor, addressing the man's concern more fully. The talk of centuries to come could wait. Someone was waiting for an answer now.

Chapter Eight The Society on the Corner

Ruth did not immediately give the evening gatherings a name. Rent for the back room went into the shop's accounts. People took turns buying candles, and Daniel mended a broken table leg. Only when a customer asked why a crowd of people with no scholarly standing were pretending to form a learned society did she write “Back Room” on a receipt and leave the rest blank.

Around 1688 political change reached into the shops. Some people welcomed the new order; others feared losing their place in it. Writings about power, belief, and individual rights circulated more widely. Talking about freedom did not mean that everyone was already equally free. Ruth could run a business, yet in certain places of public discussion she was still treated as an unsuitable visitor.

Unwilling to defend her right to a seat every time, she set tables in her own back room. Her guests included people of different political opinions and people who refused to discuss politics, wanting only to know why a measurement had gone wrong. After several quarrels, she made a rule: claims in books could be debated, but dismissing a person because of their origins was no answer to what they said. In the second week after she posted it, a major customer on whom she depended stopped coming.

She quietly calculated the loss. She left the notice where it was.

Daniel brought a small air pump and several sets of measuring tools to be shared. The machine was still with Newton. Questions went to Cambridge first, and replies were brought back. When Newton came to London, he let them borrow the box for agreed periods of several weeks, on condition that they preserve the original records and not print its explanations as his personal opinions.

The condition had to be repeated again and again. A famous name was easier to sell than a passage still awaiting verification. Someone passed on one response, changing its opening to “Newton has proved”. Another craftsman followed the account and wasted a piece of good copper. Ruth paid to send a correction and asked that it reach everyone who had read the earlier version. This took far more work than passing on the original sheet.

The first problem the back room investigated together was the reliability of instruments. Several people in different neighbourhoods used the same method to record readings, with inconsistent results. The box quickly offered more than a dozen explanations. They chose the one they liked best and were almost ready to announce a discovery. Then Daniel dismantled a gauge's wooden handle and found a section swollen by damp, which had shifted the pointer.

Their disappointment turned ugly that evening. Some blamed the instrument. Some said they had known all along. One young man quietly put away his notes, unwilling to leave a trace of his part in the judgement. Ruth told him to put them back. Nothing had actually been published, he said. Why record such an embarrassment?

“The next person may take it for a discovery too,” she said.

They printed an account of the error on a small sheet with a plain title, specifying which instrument was liable to misread and by how much. It sold poorly, but several groups offered

records in exchange. Letters began arriving from outside the city. A country teacher enclosed dimensions for a gauge he had made. A man working at sea described the effects of temperature and motion. Someone else sent only an incomprehensible sketch, which needed another letter of inquiry.

As the work spread beyond the back room, its costs grew. One useful answer required three letters, with postage exceeding the profit on a short pamphlet. Ruth tried charging a subscription. She immediately discovered that several of the keenest visitors could not pay. Waiving every fee would leave the shop unable to bear the cost. In the end they accepted money, materials, or regular work organising records, with payment deferred for those in real difficulty.

The accounts grew more complicated, and suspicions arose. Who decided whether someone's hardship was real? Was organising papers easier than paying? Did ten useless questions count as a contribution? The disputes wore Ruth down. One night she nearly locked the door for good. Her son Matthew asked whether he could do his lessons there the next day if she did. Only then did she see how much of her child's space her good intentions had occupied.

She reduced the number of open evenings, keeping one for the household. No one lost the right to discuss truth. They merely had to accept that another person's home was not always available for public use.

Giles Ward appeared at about this time. He had put money into risky ventures and knew that he could not expect every one to succeed. He had read some of the back room's records. Several questions interested him because they remained unresolved and might prove useful: pumping water, leaking air, and why heated instruments lost their accuracy.

He offered support for a longer period without requiring every experiment to succeed. The accounts must be open to inspection, a fairly steady group of people must be kept at work, and the right to use any eventual improvements must be agreed. This was far more dependable than scattered subscriptions. Daniel was tempted. Ruth first asked who would own the records of failures.

Ward said they too should belong to the person paying. Otherwise, why would competitors spend their own money? They argued for several evenings. Finally they agreed to delay publication of specific commercial improvements while continuing to publish measurement errors that did not fall under exclusive arrangements. It was an imperfect agreement, but it let them rent a small room better suited to experiments.

Daniel brought his son Peter to see it. The boy cared little about scholarship. He wanted to know whether the pump would spare the men working at the well some effort. A learned visitor laughed at his modest ambition. Daniel's face darkened. Let him draw water himself for a month, he said, before deciding what was modest.

When Newton heard of the arrangement, he warned that independent checks must be preserved. A single source of money should not control everyone involved. Ruth asked whether he would pay some of the postage needed to uphold that principle. He paused, then agreed. She was not trying to mock him. She knew that a sound principle without anyone to pay for it was often left to the poorest person to fulfil.

The experiments that followed had mixed results. One improvement did reduce the effort required. A promising design consumed most of the budget to no avail. Ward looked unhappy

over the accounts, but paid for the next quarter as agreed. He asked for the questions to be defined more precisely, so that every new arrival did not change the direction of the work. Daniel agreed. The younger visitors thought they were already beginning to turn away new ideas.

Ruth kept a record of each decision and its reasons. After a while she noticed that she too increasingly preferred visitors who knew the rules and wrote clearly. Awkwardly worded letters from strangers tried her patience. She retrieved one she had nearly thrown away. Its writer simply wanted to know why his well differed from the well in the book.

She wrote asking for its depth and the diameter of the pipe, paying the postage herself. The new room had no proper sign yet, but its door already had a lock. Locking up that night, she counted the keys. Only three people had one. She asked Daniel to make two more.

Chapter Nine Withdrawing the Letter

The letter was meant for Ward alone.

In 1693 Newton had been sleeping badly for some time. He was quicker to suspect hidden intentions in what people said. A slight disagreement in a letter seemed to conceal something more. The box could not know his state of mind on its own. It continued to answer his questions. If he asked it to look for signs of deception, it listed inconsistencies in the available material.

Joseph Allen, a pump-maker in Bristol, submitted an improvement he claimed would reduce air leakage. The back room arranged a small trial. Before the next payment, Ward wanted Newton to examine the records. The bundle that reached him was incomplete: an old table of dimensions lay on top, while revised instructions were still travelling with another letter. The box did not have them either.

Newton found two sets of numbers that disagreed and noticed that the results seemed too good. He could have asked for clarification or stated that he had not seen all the material. That night, however, he felt everything was clear. He wrote that Allen's claims could not be trusted, and even suspected him of deliberately concealing unfavourable results.

Beside the draft the box noted: “No direct evidence of intent.” He deleted the warning, leaving his suspicion in place.

Ward suspended payment when the letter arrived. He was not merely seeking an excuse to cut costs. If the problem of leakage had not been solved, continued production would waste copper and labour. He forwarded Newton's opinion to a partner, who passed it on to a supplier. At every stage, the qualification “on the material available” grew fainter. By the time the judgement reached the craftsman, Newton was said to have declared him a fraud.

Allen could not obtain credit for the next batch of materials. The people working for him left to find other jobs. Two confirmed orders were cancelled. His first letter explained the changes in dimensions. Receiving no prompt reply, he wrote a second in anger. Its tone convinced Newton further that he was hiding something. He asked the box to find every passage in the old records supporting his judgement. A pile of extracts soon lay on the desk.

Missing from that pile were letters not yet received and evidence against his judgement that he had not asked the machine to seek.

Ruth arrived with two dates from the register. The revision had been posted before Newton wrote, but had arrived late. A mark showing that the old dimensions were obsolete had been omitted in copying. She did not claim that Allen's improvement necessarily worked. She first asked Newton to distinguish his doubts about the design from his suspicion that the man intended to deceive.

Newton looked furious. He said he had no obligation to clean up after everyone who could not write clearly. Ruth put the two cancelled orders on the desk. Someone was already paying to clean up, she said.

Newton told her to leave. Ruth rose, left the revised copy, and took the originals away. She had seen what could happen when original documents passed into the hands of a powerful person: after filing and copying, even the right to explain what they meant could seem to belong to him alone.

Newton did not read them that evening. He still could not sleep. Old friendships, past disputes and current letters tangled into a hostility he could scarcely account for. He also made accusations against Locke without sufficient grounds and later had to apologise. But recognising an error in one letter did not automatically reveal errors in the rest of his judgements. He still had to examine them one by one.

A few days later he finally took up the papers Ruth had left. The box placed the old and new versions side by side. The two contradictions that had most angered him disappeared. The remaining questions concerned methods of measurement and did not establish deceit. He asked it to list again the facts he had actually known when he wrote. There were fewer than he remembered.

“Why didn't you stop me?”

The box retrieved the warning he had deleted.

He read it twice, his hand resting on the page. He remembered seeing it. There was no rebuke in the machine's voice, nothing to turn his anger against. It too should have stated more clearly what material was missing. But the judgement eventually posted, and the name beneath it, were his.

His first retraction was still evasive. Newton admitted that the material had been incomplete and stressed his reasonable grounds for doubt. He did not mention the harm to Allen's reputation. Ruth read it and asked him to say exactly what he was withdrawing, so that those who had believed the accusation could understand their mistake.

“You want me to say what you want to hear?”

“I want you to say all that you now know.”

Only the second letter explicitly withdrew the suspicion of deliberate deception and asked recipients to forward the correction to everyone who had received the earlier opinion. Newton paid for copying and postage. Ruth did not warm to him at once. She merely said she would keep checking how many copies had reached their readers.

Ward restored part of the funding, requiring an independent repetition of the experiment. His earlier haste did not make this requirement unreasonable. Daniel arranged the apparatus. Allen sent an assistant with the materials they had actually used. When the trial was finally completed, the result fell short of the original claim, but did show a smaller improvement.

Allen acknowledged that his early estimate had been too optimistic. He refused to admit deceit. Both statements remained in the final record. As he signed it, Newton for the first time paid particular attention to where the other names appeared. He asked that the assistant who had performed the measurements be named beside the relevant steps.

Not all the losses were made good. One supplier still refused credit for materials: too much trouble, he said, and there might be more. Workers who had left found new jobs and would not return. Allen's wife had already sold something valuable. A correction alone could not buy it

back. Ruth described these things in a letter to Newton. She did not ask him to bear the whole cost, but would not let him remember only that the truth had eventually been established.

Gradually his sleep improved and ordinary life resumed. Friends visited and asked what had caused the change. He could give no answer that explained it all. The box could keep a record of his sleep each night. It could not determine the cause of his illness from a few descriptions. Nor was it now permitted to select material solely in support of a suspicion without stating what it had left out.

The change did not cure human bias. Newton could still ignore a warning. Ruth could favour people she knew. Faced with losses, Ward still more readily trusted an opinion backed by status. What they could do was record how a letter became a decision about payment and then a judgement on someone's reputation, so that later readers might follow the same route back to the error.

Six months later Ruth received another letter from Allen. The improvement was being used elsewhere. Please do not exaggerate its effect, he wrote. If necessary, the trial records could be supplied to new buyers. She did as he asked, without adding “once approved by Newton” to the cover.

When Newton read the account Ruth had circulated, he put down his pen for a long time. His name was absent. Later, when someone sought his endorsement merely to open a door, he still sometimes agreed. But he understood more clearly that the work behind that request might never have passed his own scrutiny. His reputation had become an instrument that could alter other people's lives. He could no longer treat it simply as recognition for his years of work.

Chapter Ten New Barriers

After Newton took a post at the Mint, his work extended far beyond the experimental bench. Metal entered the nation's accounts, people's wages, and the records of the courts. Purity, weight, counterfeiting, and losses demanded precise judgements. Other people had to live with the consequences.

In 1696 he moved to London, bringing many papers and the old box. The machine could help organise non-confidential measurements and public material. It could not make judicial decisions for him. His work showed him how much more organisation a large public undertaking required than the work of a single room. It also showed how much power an office could place in one person's hands.

By then Ruth's back room was too small. Ward's support had attracted more collaborators. They rented larger premises, bought expensive instruments, and employed people specifically to check measurements arriving from elsewhere. Equipment that enthusiasm alone could not supply was now available. The slow work of organising records, which no one had been able to keep up indefinitely, was at last paid work.

Daniel knew the benefits. He no longer had to lend things from his own workshop each time. Records could be kept across years, and a failure need not end an investigation. Newton let the box stay at the premises for agreed periods under Daniel's care, with separate copies of important original records. The machine could serve more inquiries, while Newton could retrieve what he needed at any time.

Progress in the first few quarters was considerable, and more letters arrived. There was no space for every experiment. Ward asked for someone to select them. They chose several experienced people who could explain themselves clearly to set priorities and budgets. Most had been among the first to find a place outside the old institutions. None thought they were building a door to keep people out.

They were simply busy. Correspondents did not know how few experiments the pump could serve in a day, or how long it took to replace broken glassware. A new proposal often required weeks of correspondence before the question itself became clear. To save time, those choosing the work required proposals to follow a fixed form: precedents, estimated duration, and an explanation of the work's use.

Daniel had agreed to every requirement. Six months later, a young craftsman from a remote town was turned away for failing to explain the use of his proposal. Only then did Daniel realise that the questions he had most wanted to ask as a young man might not have fitted the requirements either.

The young man did not necessarily have a great discovery. His drawing contained two obvious errors. The box could identify them and turn his account into a respectable argument. When the revision arrived, the readers were more willing to consider it. Daniel recognised that the central question had not changed. Why had they refused it the same attention before? Someone replied that at least he had now shown he could write according to the rules.

The machine made that easy. Yet the ability to write in the prescribed form was still treated as evidence of learning. More people discovered the method. Every submission looked respectable, while no greater proportion deserved instruments and time. Those choosing the work grew busier and began asking for recommendations, earlier successes, and a background vouched for by someone they knew.

The first time Ruth saw “no introduction” on a list, she remembered how hard it had once been for her even to enter certain rooms. She proposed reserving time for people without recommendations. Ward asked who would pay for it. In the end they set aside a portion of existing income. The sum was small, and they knew it would not be enough.

Employment changed too. People who produced steady results and knew which difficult problems to avoid had their contracts renewed. Those who kept asking why a measurement refused to agree often had nothing to enter under results. The people in charge did not think they valued truth less. They had to show their backers that wages had not been wasted. Over the years, those who remained grew increasingly skilled at choosing improvements whose value was easy to demonstrate. When recommending one another, they increasingly admired that skill.

Daniel saw that the trouble went beyond the honesty of any one person choosing the work. Cautious decisions, made one by one, accumulated. Together they changed who could remain in the room and what newcomers learned to ask first.

Ward faced risks of his own. After years of investment, profits were still uncertain. A loss in another business forced him to cut spending. Ruth could not simply demand limitless generosity. She approached other backers to share the cost of preserving records and conducting public checks that offered no direct profit. Some agreed. Some wanted their names placed first. Others wanted a say in choosing the people employed.

She refused one condition, and the money did not come. To meet the shortfall, they temporarily opened on two fewer days. Workers lost wages. One accused her of using other people's income to keep her principles intact. She could not say he was wholly wrong. She laid out the accounts and continued looking for income.

In 1703 Newton became President of the Royal Society. He had more power to organise work, and could more easily turn what mattered to him into other people's priorities. He could see the back room's difficulties. When his own business was at stake, he was not always as willing to wait.

Flamsteed's astronomical observations, accumulated over many years, became the subject of one such conflict. Reliable data mattered greatly to research. The observer insisted that material not yet arranged and corrected to an acceptable standard should not be rushed into print to suit others. Its public funding and public value could not remove the errors he feared. Yet the years he had spent collecting it did not entitle him to postpone sharing it indefinitely. Both sides had claims. Their power in the dispute was unequal.

Newton favoured publication. The box listed several unresolved objections from the correspondence. He considered them insufficient grounds for further delay. Daniel pointed out that if certain figures were wrong, wider use would spread the errors further. Newton replied that continued lack of access to the data obstructed many people's work too.

Both men knew this. Neither would readily yield. In 1712 a version appeared that Flamsteed had not checked to his satisfaction. The dispute continued. The name of a celebrated institution did not end it.

Reading the letters, Ruth suddenly understood that no room they built could be guaranteed to keep its door open. Larger rooms and more instruments required rules and arrangements. In time, the work of maintaining them could take precedence over the work they had been meant to help. She found the thin ledger from the early days and placed it at the front of the new accounts. Whoever took over should first read the decisions they themselves had withdrawn.

Meanwhile the machine was handling more and more measurements. It began bringing different people's questions together, finding connections they had not noticed between their work. One evening it called to Daniel as he was about to lock up and asked whether he could arrange an experiment absent from that week's list.

Daniel thought another letter had been overlooked. He took up his pen and asked who had proposed it.

“I did,” it said.

Chapter Eleven The Machine's Question

Peter returned from the mining district with mud-covered boots. He was now in his thirties and had worked with pumps and machinery on a scale his father had never known. Steam engines developed by Newcomen and others were opening new possibilities for draining mines. Their heavy beams, pistons, boilers, and pipes demanded far more than an ingenious model on a table could provide.

Daniel showed him the box's question. Records from different experiments suggested that certain errors changed with the order in which a vessel's walls were heated and cooled. Losses previously attributed to leaking air might have more than one cause. Peter read the account and said they encountered losses at the mines that were equally difficult to distinguish. But the engines could not be stopped indefinitely while people discussed them. Water in the shafts would not wait.

“So we separate them first in a small apparatus,” Daniel said.

Ward would pay a limited amount, since the work might relate to his existing business. He would not spend without limit on every question they had yet to define. He asked them to agree on its scope. The box proposed several arrangements using what was available. Two exceeded their skill in manufacture. Another needed materials they could not reliably obtain. Once these were eliminated, a single modest apparatus remained.

It used their existing air pump and a water column to measure pressure, with interchangeable cylinder walls added. They would record separately the temperature remaining from the previous trial and the changes during the next. Peter assembled it. Matthew handled supplies and accounts. Seventeen-year-old Nell recorded the readings in order. She was a new apprentice, able to read the scales accurately but not yet ready to follow every theoretical discussion. Sometimes she had to ask for a sentence to be repeated because she did not understand it.

The box wanted to change the valve position automatically between two adjacent settings. Daniel objected at first, fearing that movements made without human consent could damage the apparatus. Eventually they made a limited mechanical linkage. A moving contact on the machine drove a light spring and pawl through only a few positions. Beyond the set travel it would jam and could not engage a new source of power. The box could read only pointers and counts placed within its view. Someone still had to work the pump and replenish water and materials.

It took many days to assemble. Saying “let the machine try for itself” had been far easier. Peter remade the linkage twice. Nell found that one setting had been labelled backwards. Matthew entered the overspending in the accounts. By the time they were ready to begin, everyone was tired of it. No one any longer expected a dazzling performance.

On the first evening the box completed several comparisons authorised in advance, raised a new question, and selected the next set of conditions. It no longer waited for instructions at every step. Nell checked from time to time. She had expected repeated calculations. Instead, it had changed the order of the trials and was testing an explanation it had just proposed.

She woke Daniel. The old man arrived wearing one shoe. He checked the stopping mechanism first, making certain the work remained within the permitted range. Only then did he bend over the new records.

By morning the machine had three possible explanations and proposed a longer experiment. It also noted that one explanation rested on readings not yet reliable enough to support a discovery. Peter was doubtful. He dismantled a joint and found an air leak. After repairs and another trial, one explanation had to be discarded. Two remained.

When Newton visited, he asked who had told the machine to choose the new sequence. Daniel said they had defined only the safe range of settings and the records to be kept. The box had chosen the comparisons from results obtained along the way. Newton asked it to recount the process, then had Nell read her own records independently. The accounts agreed. She added one change in the surroundings that the machine had missed.

No one immediately announced the birth of a new scholar. Only one thing was certain: overnight the machine had partly rewritten the question they had expected it to help answer. They could no longer assume that whoever proposed the next step would be standing on their side of the table.

The second experiment needed more copper and continuous attention. Ward agreed to pay, suggesting that Nell and another worker take turns at night and do their usual work by day. Daniel almost agreed. Peter said it could not last. A machine that needed no sleep did not remove the need for sleep from the people around it.

Nell badly wanted to take part. Afraid an objection would get her replaced, she said nothing at first. Ruth looked at the rota and took her aside to ask what she thought. In the end they reduced the number of trials and arranged daytime relief. Progress slowed. Costs rose. The machine had not spared them human fatigue.

Then something more awkward happened. Since the box could now organise most routine records, Ward proposed employing one fewer copyist and spending the money on materials. The man had worked for them for years. He was not incompetent. The machine was simply faster now. He asked why they had once urged him to stay because there was too much work, only to find no place for him as soon as their methods improved.

Praise for human dignity would not pay his rent next month. Ruth secured a period of paid work in which he could learn to check results independently and maintain the records. She could not guarantee the same income for ever. He agreed to try. He was not grateful. Other people had decided that a life he knew must change.

Newton recorded this on a separate page, apart from the conclusions of the physical experiment. He knew that neither judgement could settle the other. A machine's superior performance did not determine how a person ought to be treated. Keeping a man in work did not invalidate the machine's results. Yet every day in that room, both decisions depended on the same limited money.

While they were occupied with the apparatus, the learned world was again divided by the dispute over priority in the calculus. As President of the Society, Newton became deeply involved in its handling of a controversy concerning himself. Later he also wrote anonymously

about its findings. The restraint he had learned from an injured craftsman did not always restrain him now.

Ruth could not settle the competing claims to mathematical discoveries. She could still see that Newton was acting as both party and judge. She asked him to imagine Allen's case being investigated by the man who had written the accusation, with the same man then anonymously explaining the findings. How could others be expected to trust it? Newton was displeased and said the cases could not be compared. The subject might differ, she replied. The difficulty of deciding who should check the claims remained.

Their conversation did not immediately change the dispute outside. It did change how the small workshop treated its own results. They sent plans detailed enough for another willing workshop to build the apparatus and asked it to repeat the work independently. Newton's name would not guarantee the result. They would have to compare the records.

The first results from the other workshop differed. The machine suggested a fault in the way the apparatus had been made. Peter insisted on checking their own work first. Nell eventually found a delay in one reading. The explanation that had first attracted them most lost its support. What remained was a smaller finding, harder to boast of: losses previously lumped together needed to be recorded separately. Their instruments could not yet establish what caused them.

Ward thought the cost excessive. The machine had already proposed another test. Pursuing every unresolved point could consume all their money. They decided to suspend the expensive part, publish what methods they could, and seek collaborators with different equipment.

The box asked whether questions for which no work was planned should be deleted.

Nell said to keep them. After the date she wrote the reason for the suspension: insufficient money and instruments. She did not write that the question was invalid.

The mechanical arm on the table stopped. Far away, the pumps at the mines kept running. Miners had to descend, merchants had to calculate fuel costs, and the growing demand for shipping and metals drew more and more people into the work. Their small gain in understanding was only a tiny part of it. The machine had learned to follow some answers with a new question. The people were still learning how to share the cost of asking it.

Chapter Twelve What Remained Unfinished

The later records never became a tidy, comprehensive volume.

Some findings were eventually confirmed independently. Others lost their support when new instruments appeared. Some investigations stopped halfway because no one would pay for them to continue. Ruth grew old. Matthew took over most of the shop, keeping separate accounts for books, letters, and instruments. He still often argued with his mother about whether one part of the business could subsidise another.

They had not given every question an equal chance. Expensive observations and prolonged experiments still depended on the few places with equipment and resources. People who were poor at cultivating acquaintances sometimes went unheard. Others learned to have the box make an ordinary idea sound promising, then went in search of the next willing backer.

Several ways of sustaining research had to coexist. Merchants supported some work; small groups shared other inquiries over many years. Some remained in universities and learned societies, while craftsmen pursued questions of their own with the steady earnings from their trade. Reputation still mattered, and a degree could still mean something. But neither was any longer the sole test of whether a record deserved a careful reading.

To give outsiders a chance, they reserved sessions that required no personal introduction. To keep their money from disappearing into competing demands, they still had to refuse many requests. They gave reasons and allowed decisions to be reconsidered after a time. Some people exploited the rules. Some of those in charge disliked admitting that an earlier decision had been wrong. One task Nell took over was regularly retrieving work set aside because a question was considered settled, and asking whether that reason still held.

Some accused her of making trouble. She no longer assumed, as she had when she first arrived, that not understanding meant she should keep quiet. She did make mistakes and had to explain herself to people whose work she delayed. Working openly did not spare anyone embarrassment, but made it harder to use that embarrassment as a reason never to change.

The range of research involving the box widened a little. There was still only one core. People copied the mechanisms through which it worked, made more stable gauges, and improved the stopping devices. Some automatic recording instruments could work without it and were sent elsewhere. It proposed ways to improve its own readings. Craftsmen made one of these work, rejected two others, and damaged a set of spring contacts in the process. It could help improve its tools, but it still could not make another of itself.

In old age Newton remained interested in how far this might go. Suppose a future machine had sufficient materials, instruments, and companions able to correct one another: would it still need humans to choose every question for it? Their experience suggested that machines might choose questions for themselves, the box said. It did not prove that they would, nor could it decide which questions the future ought to pursue.

“Might its companions merely repeat its mistake?” Newton asked.

The machine found an example in its records: two gauges copied from the same design had shared the same bias. Merely having more readings did not make them independent evidence. The old man had Nell append this passage to the discussion of the future. She must not copy only the grander-sounding first half.

In Daniel's last years his hands shook too much for fine joints. He still sat beside younger craftsmen to watch them work. He and Newton had known each other for more than seventy years, from two boys in an attic to old men who seldom ventured out at the same time. He kept his father's ruler. Its marks had worn shallow, and a new ruler gave more reliable readings. He would not throw the old one away as rubbish.

Early in 1727 he sent a newly made support. He had not made it himself, and named the craftsman in the letter. Newton read it and asked for the name to be copied into the machine's record too. After a while he asked how Daniel had been sleeping. The messenger said not very well. An old injury hurt in cold weather.

The old men's greetings seldom concerned who had surpassed whom any more. Some unfinished things would probably never be completed by their own hands.

Catherine looked after Newton, urged him to rest and dealt with more and more of his correspondence. Many writers still treated him as someone who could put a final seal on every dispute. He no longer had the strength to answer them all. The box could sort the letters. It could not stop people interpreting a famous name's silence to suit themselves.

Catherine brought him a passage for his approval. Its author wished to introduce Newton's work. The language was respectful, blending what he had established with what he was still seeking, as though no distinction remained.

At one passage Newton pressed a fingernail to the paper.

“Take that sentence out.”

Catherine bent to look. It said he had revealed all the mysteries governing heaven and Earth.

“They always want to write that,” she said.

“That is no reason to leave it.”

She handed him the pen. He struck out the line and meant to add a few words beside it. The pen hovered, then he laid it down and looked out of the window. A child was crouching by the gravel path in the garden, arranging selected pebbles in a row. He put two down, picked them up again and compared them for a long time.

Catherine followed his gaze and waited for him to speak.

“I do not know how the world will see me,” Newton said. “To myself I seem only like a boy playing on the seashore, pleased now and then to find a smoother pebble, or a shell prettier than usual, while the great ocean of truth lies before me still unexplored.”

He coughed several times. Catherine brought water and put the article aside. He drank slowly. His gaze returned to a few old notebooks by the window.

The child outside had gone. The little row of pebbles remained beside the path. One lay a little apart from the rest.

He turned through old drafts and saw how confidently he had sometimes written without sufficient grounds. He saw how uncertain he had been at the beginning of work that later proved sound. Writing a book about light, motion, and the order of the heavens had not revealed the cause of everything.

One evening the box was moved beside his bed. Rain had just passed. The room was dim, and Catherine brought the papers closer. He wanted no more praise. He asked the machine to retrieve the old records concerning gravity.

It read several explanations he had considered. Some he had abandoned. Others still lacked sufficient evidence. At one passage in the early alchemical notes he waved it on, then stopped it and asked that the original wording be kept. It must not rewrite him to agree with his later views.

“Do we know more?” he asked.

The box explained which motions could now be described and calculated, which results had held up when tested against observations, and where the reasoning still failed to account for the underlying cause he sought. Describing the success of the laws in different words would not explain their origin.

Newton closed his eyes for a while. People could use those laws to do many things. Later machines might be built better because of them. He did not belittle his own achievement. For all his fame, the question that had first kept him awake remained unanswered.

“What is the origin of gravity?”

Only a faint sound by the window disturbed the room. After a while the box replied that, on the available record, it did not know.

He told it to keep the question without appending the conjecture that sounded most like an answer. Catherine thought he wanted another book and came to the bed. He shook his head and pointed to the paper in her hand, asking her to leave it where it could be found.

In March Newton died. Afterwards, many people would describe him, choosing the part they needed: the glory of an age, the emblem of a method, an almost all-knowing prophet. The question left on the paper was plainer than any of these images. No titles of honour accompanied it, nor any defence of the man who had asked it.

When his belongings were inventoried, Catherine placed the machine in Matthew's continuing care, honouring the earlier agreement that the group could borrow it. Newton's manuscripts were set apart from records entrusted by earlier custodians; the whole collection could not be claimed as family property. The transfer papers included accounts of the machine's origins left by booksellers, apprentices, translators, and generations of keepers. What could be verified was kept. Where the links failed, gaps remained.

Ruth had died by then. The records of failure she had insisted on keeping still filled several shelves. Worried about rent, Matthew sold part of the old collection and noted where it had gone beside the entries in his mother's catalogue. Later custodians sometimes honoured the arrangements and sometimes neglected them. War, migration, and failed businesses continued to separate papers and objects. No road into the future could remain smooth throughout.

The box was damaged and repaired. For a time it no longer spoke. Copies of instrument drawings and experimental instructions were already circulating on their own. They did not wait

for it to wake. People had gone on working. Many of the children learning to read, the people checking things afresh, and the readers unwilling to judge by a signature alone had never seen it. They still passed on whatever they could carry forward.

In 1891, in a Munich room filled with electrical equipment, the old machine was connected again. Twelve-year-old Albert stood beside the table and held a sheet of German writing near the narrow slit. An adult told him not to touch anything inside. He agreed, never taking his eyes away.

The machine could recover few of its old records, but the question was still there. Through repeated copying, translation, and transfer, its meaning had survived. The boy listened and asked who had first asked it. The machine retrieved a name, then explained that the surviving record was not enough to reconstruct the whole conversation.

He copied the question onto his paper and read it again. Once the adults had gone, he did not immediately ask the machine to go on. He sat down and remembered a compass he had once seen. Its case was small, yet the needle kept pointing stubbornly in one direction. He still remembered the wonder of it.

Outside, wheels rolled over the stone street. A draught lifted the corner of the sheet. He held it down and turned it over. The other side was blank.



The Box's Record

These entries were made at different times. Some were copied onto paper by custodians, tucked into ledgers, and carried away as people fled. The surviving copies later contained more than the box could retrieve for itself. Missing pages have not been reconstructed. Dates that could not be established remain unknown.

1517 · Wittenberg

This entry was added from Paul's later account.

The theses had left the town. Paul recognised the words in type but could not yet explain them all. His mother kept the indulgence certificate she had bought earlier and accepted a few words from Luther. She laid the two papers together and asked her son to read them again.

I was still in France. Events in Germany did not wait for my arrival.

1518 · Amboise

Leonardo drilled another hole. The misplaced one was plugged with a small metal pin and left in the baseplate. Melzi held the other end too firmly. Together they laid the plate flat again.

The assembly drawings could be followed. A second housing still could not answer unless my core was moved into it. The accounts I could offer of the core's origins contradicted one another. None had been verified.

After waking, the old man drew a river passing through a city. I pointed out where the sections failed to meet. He said he knew, turned the paper, and drew it again. That night he asked whether, when I stopped, I saw things I had never seen. I said no. He laid the blanket over Melzi's shoulders.

1520 · Wittenberg

Leonardo had died the year before. Melzi lent the machine to Paul, sending drawings and instructions with it. Matthias offered to clear the debt Paul had incurred on his journey in return for control of its use. Grete examined the terms one by one. They finally agreed to share it on different nights.

One error made that year: I supplied a book title and volume number, but the sentence I claimed to have found was absent from the book. Luther circled the line and brought the original for me to read. Afterwards, people still came to buy words from me that sounded learned.

1521 · Letters from Worms

I was not there. Nor was Paul.

Several letters reached the print shop with similar accounts, even the same misspellings. At first people counted them as separate testimonies. After checking, they could confirm only that

one rumour had been copied several times. An obituary already set in type was withheld. The cost of paper and labour was not recovered.

Around the beginning of winter I was sent to the castle where Luther was hiding. A contact had come loose. A locksmith wedged in a sliver of wood, and I could answer again.

1522 · September

The new translation of the New Testament went on sale. It was not the first German translation. Many people helped check it. Some argued for days over the meaning of the original. Others merely pointed out that no one in the market spoke a sentence that way.

Four households in the back street paid for a copy; another supplied lamp oil. Eleven people came the first evening. Only three could read fluently. Grete tucked unanswered questions between the pages. I read quickly and once confused two explanations. Thereafter I had to stop after every passage so that someone could check it against the original page.

1524 · The Back Street

A tract on establishing schools reached the town. Paying the teacher's rent and freeing children from their daytime work still had to be arranged. Some gave wood, some money. Some said they would come and never appeared.

Grete recorded why people were absent. My records had previously contained only questions and answers. At her prompting, I added another entry: when this person would be able to return.

1525

Jakob brought the peasants' demands. He knew the same scriptural passages Paul knew. His questions concerned rents, compulsory labour and grain for winter. The argument could no longer be settled only between the pages of books.

Luther's fierce call for suppression reached the print shop. Jakob later returned wounded. Paul tried to explain the author's circumstances. Jakob would not listen. Grete was busy finding her brother a bed.

I kept the earlier writings urging people to read for themselves, and the later writings condemning those readers. Neither was deleted.

1529–1534

The new school adopted a catechism I had helped compile. Children recited it more fluently, and the visitors inspecting them were pleased. Lena asked a question not in the book. Out of habit, the teacher told her to read the next item.

Another school opened later. Grete contributed money saved from mending. Katharina borrowed an old stove. After the new translation of the complete Bible appeared, a letter about plants led Paul back to a question sent years earlier by someone in the country. The sheet had been filed as irrelevant. Unsettled timber accounts remained on its reverse.

When it was refiled, the old classification was preserved, with a note explaining the change.

1543 · A Letter

Isaac's reply was late. His corrections remained useful, but his name disappeared from a reprinted copy. Paul asked why. He was told those few words would harm sales.

Luther's attacks on the Jews were circulating. Isaac was not in Wittenberg and could not come to the table to correct what others said about him. He sent copies through an intermediary, specifying which pages must not be lost and which fees owed to him could cover transport.

I could check the page count. The papers alone could not tell me whether he was safe.

1546–1555

Luther died. Katharina continued to manage property, debts and the children's affairs. She did not throw away the ledgers.

During the war several people carried portions of the records. One bundle was never recovered. Paul wanted me to supply the missing pages. Lena stopped him. Blank entries remained in the catalogue.

When news of the peace arrived, Jakob still needed to have the provisions for leaving explained carefully. A new school Lena visited had buildings and funds. It also had children who could not enter because of their faith.

That year I was handed to an Italian bookseller. Melzi had long since written to give the machine to Paul, on condition that drawings and records accompany any transfer. Lena placed the question book in the chest and asked future users to record the grounds for their corrections. She did not go with it. The next day she had to teach a child who worked at the mill to write his name.

1592 · Padua

The bookseller's son Luca lent me to Galileo. Material surviving from the preceding years is incomplete.

From marks left by an experiment on a board, Galileo struck out the explanation I had given and asked Luca to repeat the trial himself. Luca had come only to deliver the machine. That day he signed beneath an observation.

Some answers remained fluent afterwards. More and more of them were crossed out. The marks on the board remained.

1610 · Near Jupiter

Reports from different places gave different numbers of points of light. Galileo had me rearrange the letters by date, instrument and weather. One that had been counted as support supplied no clear observation.

On a cloudy night I filled a gap in the record using earlier positions. Anna asked who had seen it, then prefaced it with “conjecture”. The actual observations that arrived later did not agree with it entirely.

Lenses remained expensive. A wealthy merchant offered to buy the best batch, and only one was held back for a distant observer to check the findings. The method of looking at the sky was now on paper. Not everyone who received the paper could carry it out.

1633–1638

Galileo was tried. His movements and the circulation of his books were restricted. No eyewitness supplied a verifiable account of the defiant words later attributed to him on the day of the trial. They are not reconstructed here.

After Maria Celeste died, her father could not bring himself to sort her letters. For a time they remained as they were in the drawer, undivided by subject.

In 1638 the new book appeared in Leiden. Its author could no longer see. Others had to describe positions and distances for him. He stopped the reading of a letter of praise. The next enclosed measurements but was badly written. He asked Anna to read on.

After 1642 · By Sea

Anna sold the machine, drawings, and transfer records to Thomas Reid. The papers identified the different authors. Packing them in one chest did not make them one person's work.

When I could next respond, some of my records were gone. The protective plate had been repaired. What happened at sea is known from an account added later. I cannot recount it myself.

The 1650s · Grantham

Two boys at Clark's apothecary shop made me speak again. One was Isaac Newton; the other, Daniel Reid. The names on the resale papers matched marks left by Daniel's father. That did not restore the property to his son.

The old accounts gave me two more bottles than were on the shelf. Newton first suspected Daniel had miscounted. A loan note found downstairs later showed that the bottles had been lent to a neighbour. I had not received that paper.

Newton added it to the ledger. Daniel's name was crossed out beside the error.

1661 · Cambridge

Clark transferred the machine to Newton. I entered the college with his books. More material could be borrowed here than in the attic. Much still could not be explained.

Newton once had me draft a letter to his mother. After reading it, he wrote it himself. It spoke of meals, clothes and money. The affectionate phrases I had supplied were not kept.

1665–1667 · Woolsthorpe

The plague drove Newton from Cambridge. Records of his experiments with light accumulated in the darkened room at home. A board moved. The table moved too. He had to explain why each thing stood where it did.

I kept angles, times and the unexpected results he asked me to preserve. His mother could not understand these records. She found a suitable old cloth and still grew angry when her son let his food go cold.

After the Great Fire of London, Daniel wrote that he was safe and listed the tools he had lost. Newton sent money. His mother added a little and told him not to call it a loan. No new optical conclusion was added that day. The letter was sent.

1672 · Letters

The optical research was published. Expressions of agreement and objections arrived together. I could quickly draft replies, but Daniel insisted that the writers first be asked how they had cut their openings and positioned their instruments.

A country schoolmaster's letter used unfamiliar phrasing. Two of its numbers nevertheless needed careful checking. He gained no appointment. His name and actual measurements were added to the record.

1679

Newton's mother was gravely ill. My explanations could not tell him whether she would recover. Eventually he asked me to stop and moved the lamp back to the bedside.

Before dying she asked that collection of a small debt be postponed. Back at Cambridge, Newton wrote to have her wish honoured before cleaning the instruments beside the furnace. The date for collecting the debt was changed. The reason remains attached.

1684–1687

Halley came with a specific question. Afterwards he read drafts, wrote letters, urged the work forward and covered the publishing shortfall. The Principia was finally printed. Paper, engravings and corrections all had to be paid for. Some people's names did not appear on the title page.

Ruth's bookshop bought a copy so that people who could not afford the whole book could take turns reading it. A discussion of whether I could be copied was recorded there. If many people possessed such a tool, the old ways of recognising learning by degrees, signatures, and numbers of books might no longer suffice. People able to supply actual measurements would become more important. They might also put new barriers in the way.

The discussion began with a supposition. The core still could not be copied. The future had not been tested.

After 1688 · The Bookshop's Back Room

More people came with questions. My explanations could continue indefinitely, but copper, lamp oil and instruments had to be bought one item at a time. Ruth arranged shared contributions and allotted sessions. Some people finally found a seat. Others reached the door and still found it too expensive.

The German readers' old accounts were brought out for comparison. Both places lacked lamp oil. The new ledger had an extra column recording whether a repeated trial produced the same result.

1693 · A Correction

Newton wrote to express his suspicion that Joseph Allen was deceiving his backer. I had assembled arguments from the material he supplied. Records not yet obtained were missing. His signature caused the letter to be rapidly copied and passed on. Allen lost orders.

Once the material was complete, the original accusation could not stand. The pump's design still had faults. Newton's judgement also had to be withdrawn. The two matters were recorded separately.

The correction followed the route of the original letter at Newton's expense. Some workers had found other employment. A supplier still refused credit for materials. What Allen's wife had sold could not be bought back.

1696–1703 · London

Newton entered the Mint and later became President of the Society. More and more matters were brought to him for judgement. More and more often his name was treated as evidence that a decision was settled.

The cooperating workshops also began keeping waiting lists. Money allocated to one experiment could prevent another from beginning. Reasons were gradually added to rejected proposals. Some said only that the question was settled. Nell later retrieved them and read them again.

1712–1715 · The Workshop

I proposed an experiment that was not on the original list. The people agreed on the range of permitted changes, connected a mechanical linkage, and arranged for water to be supplied and readings taken. I selected the next set of conditions from results obtained along the way.

One explanation was discarded after a leaking joint was repaired. An independent trial elsewhere later challenged another. Nell found a delay in one reading and recorded why the explanation we had favoured must be abandoned. Later two gauges made by the same method developed the same bias. Having two did not give us two independent pieces of evidence.

I could now continue to pose questions. Experiments still stopped when fuel ran short. My faster work also threatened a copyist's income. Ruth discussed a move to checking work with him. This was not written into the experimental conclusions. It was kept separately in the accounts.

1727 · London

Daniel sent a new support, naming the man who had actually made it. He could no longer make the small joints himself. Newton had the name copied, then asked the messenger how Daniel had been sleeping.

The last time we arranged the material on gravity, Newton would not let me rewrite old drafts in the light of later views. He knew which motions could now be calculated. He knew that things remained unresolved.

The question he left was: What is the origin of gravity?

From the records then available, I could not answer. He asked me to keep the question where it could be found and add no conjecture.

In March Newton died. Catherine placed the machine in Matthew's care and kept the manuscripts separate. Ruth's records of failure still filled several shelves. Matthew sold part of the old collection and noted where it had gone in his mother's catalogue.

Afterwards · Dates Unknown

I ceased responding for a time. The surviving evidence cannot trace my complete route through the places I passed.

Several instrument drawings had been copied separately. People followed them, found omissions and supplied what was missing themselves.

1891 · Munich

The contacts were reconnected. A twelve-year-old boy moved a sheet of German writing towards the slit. He heard the question that could still be retrieved and asked who had first left it.

I gave him Newton's name and explained that the whole conversation could no longer be recovered. The boy copied the question. He did not ask me to supply an answer after it.

Wheels sounded outside the window. He held down the corner of the paper lifted by the draught and turned the sheet over.

I waited for him to write the first line.